

SENIOR MATHS

FINANCE & MONEY

Kevin Dunne



Quality ready-to-use
resources



SENIOR MATHS

FINANCE & MONEY



Kevin Dunne



Kevin Dunne is a highly experienced secondary school maths teacher and a valued member of the Instant Lessons team of authors. He has taught for two years in primary schools and 32 years in secondary schools in New South Wales. He also taught for a year in New Zealand as an exchange teacher. Kevin has been a member of the NSW Senior Mathematics Syllabus Committee and the HSC Examination Committee. He marked HSC examination papers for 10 years. This is the fourth book he has written in this series of *Assignments in Senior Mathematics*.

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I would like to thank my wife, Rikki, for her invaluable assistance in the preparation of this book.

Kevin Dunne

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introduction

This series has been written to provide comprehensive coverage of selected topics from the senior mathematics syllabus. This book concentrates on Financial Mathematics. Most topics are aimed at the senior syllabus, but some may be useful in junior courses. Each worksheet has been carefully graded with the more difficult questions towards the end of each set. Where appropriate, exercises include a worked example to show the most common methods of working. Students should be able to complete the worksheets with minimal help from the teacher. Each topic has sufficient questions to occupy students for at least a 40-minute lessons.

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A **salary** is a fixed payment made in return for work. It is often quoted as an amount per year. People earning a salary may or may not be paid overtime. Many professionals, teachers, nurses and government employees are paid a salary. Salaries are generally paid weekly, fortnightly or monthly. The following can be used for salary calculation:

1 Year = 365 days or 52 weeks or 26 fortnights or 12 months

Note: Some of these are not exact. They are used for ease of calculation. **One month is not four weeks.**

e.g.

- 1** David earns a salary of \$56 000. Determine his fortnightly pay.

$$\begin{aligned}\text{Fortnightly pay} &= \$ (56\,000 \div 26) \\ &= \$2153.85\end{aligned}$$

- 2** Sarah has a weekly salary of \$674. Her employer decides to pay her monthly instead. Determine her new monthly salary.
Note: One month is not four weeks.

$$\begin{aligned}\text{Annual salary} &= \$ (674 \times 52) \\ &= \$35\,048\end{aligned}$$

$$\begin{aligned}\text{Monthly salary} &= \$ (35\,048 \div 12) \\ &= \$2920.67\end{aligned}$$

- 3** Roger's salary is paid monthly. Ahmed's salary is paid weekly. Roger is paid \$2145 and Ahmed is paid \$490. Which is the greater annual salary? By how much is it greater?

$$\begin{aligned}\text{Roger's salary} &= \$ (2145 \times 12) \\ &= \$25\,740\end{aligned}$$

$$\begin{aligned}\text{Ahmed's salary} &= \$ (490 \times 52) \\ &= \$25\,480\end{aligned}$$

Roger's salary is greater than Ahmed's by \$260 a year.

For each of the following weekly pays, calculate the corresponding annual salary.

1 \$502

2 \$756

3 \$958

For each of the following fortnightly pays, calculate the corresponding annual salary.

4 \$896

5 \$998

6 \$1758

For each of the following monthly salaries, determine the corresponding annual salary.

7 \$2145

8 \$3168

9 \$5209

For each of the following weekly pays, calculate the corresponding monthly pay.

10 \$347

11 \$896

12 \$450

13 George receives his salary weekly. If he is paid \$1520, what is his annual salary?

14 Sally's fortnightly pay is \$1400. Calculate her monthly salary.

15 Carlos receives an annual salary of \$54 000. Calculate his monthly salary.

16 Amelie's annual salary is \$62 972. Determine her

i) weekly pay

ii) fortnightly pay

iii) monthly pay

17 Andre earns \$1056 per week. Determine his monthly pay.

18 Your employer wishes to pay you \$455 a week. How much would this be as a monthly payment?

19 Lisa receives \$4150 per fortnight. Determine her monthly pay.

Determine which is the better annual salary and by how much.

20 i) \$867 per week

ii) \$46 592 a year

21 i) \$963 per fortnight

ii) \$24 960 a year

22 i) \$1265 per fortnight

ii) \$2565 per month

Determine which is the better weekly pay and by how much.

23 i) \$870 per week

ii) \$44 668 per year

24 i) \$1950 per fortnight

ii) \$4200 per month

25 i) \$3471 per fortnight

ii) \$1610 per week

A **wage** is based on an hourly rate of pay. It is usually paid weekly or fortnightly.

e.g.

- 1** Freddie is paid \$14.50 per hour. If he works a 35-hour week, calculate his weekly wage.

$$\text{Wage} = \$ (14.50 \times 35) \\ = \$507.50$$

- 2** Dianne works a 38-hour week for \$414.20. Determine her hourly rate and her yearly income.

$$\begin{aligned} \text{Hourly rate} &= \$ (414.20 \div 38) \\ &= \$10.90 \text{ per hr} \\ \text{Yearly income} &= \$ (414.20 \times 52) \\ &= \$21\,538.40 \end{aligned}$$

- 3** Jim works a 5-day week, 7.30 am to 3.15 pm. His hourly rate is \$19.60. What is his weekly wage?

$$\begin{aligned} \text{Daily hours worked} \\ &= 7 \text{ hours } 45 \text{ minutes (i.e. } 7\frac{3}{4} \text{ hours)} \end{aligned}$$

$$\begin{aligned} \text{Weekly wage} &= \$ (19.60 \times 5 \times 7\frac{3}{4}) \\ &= \$759.50 \end{aligned}$$

-
- 1** James, a waiter, is paid \$10.50 per hour. Determine his weekly wage if he works $37\frac{1}{2}$ hours in a week.
- 2** Monica is paid \$9.40 an hour. What is her weekly wage if she works a 37-hour week?
- 3** Albert's hourly rate is \$11.20. Calculate his weekly wage if he works the following hours:
- i) 30 hours ii) 41 hours iii) 36 hours, 30 minutes
- 4** Delta works from 8.00 am to 5.00 pm, Monday to Friday. She has an unpaid break of 45 minutes. If her hourly rate is \$16.40, calculate:
- i) her daily wage ii) her weekly wage iii) her annual wage
- 5** Abbi is paid \$21.60 per hour. She works from 6.00 am to 3.15 pm, Monday to Friday, with an unpaid lunch break of 1 hour. Determine:
- i) her daily wage ii) her weekly wage iii) her annual wage
- 6** Sahil works in a cake shop from 8.00 am to 4.15 pm, 5 days a week, at an hourly rate of \$7.80. He is not paid for his lunch break of 45 minutes. What is his:
- i) daily wage? ii) weekly wage? iii) annual income?
- 7** Complete the following time sheet and calculate Harem's weekly wage, if his hourly rate is \$10.80.

Day	Start	Finish	Hours worked
Mon	7.00 am	3.00 pm	
Tues	8.00 am	3.00 pm	
Wed	7.00 am	2.15 pm	
Thu	8.00 am	5.00 pm	
Fri	7.00 am	2.30 pm	

- 8** Steve works a 38-hour week. His weekly wage is \$431.30. Calculate his hourly rate.
- 9** Irene works in an office for $37\frac{1}{2}$ hours per week. Determine her hourly rate if her weekly wage is \$326.25.
- 10** Wendell works $35\frac{1}{2}$ hours a week. If he earns \$308.85, what is his hourly rate?
- 11** Amber works 39 hours a week. Determine her annual salary if she is paid an hourly rate of \$13.50.
- 12** Alby's hourly rate is \$9.50 and he works a $37\frac{1}{2}$ -hour week. Determine his annual income.
- 13** Malcolm's award rate is \$11.30 an hour. He needs to earn \$412.45 a week. How many hours must he work?
- 14** Linda's yearly income is \$18 969.60. What is her hourly rate if she works a 38-hour week?
- 15** Garth has an annual income of \$21 060. What is his hourly rate if he works a $37\frac{1}{2}$ -hour week?

Many workers receive **leave or holiday loading**. This is usually an extra $17\frac{1}{2}\%$ of their 4 weeks holiday pay.

Overtime is paid when a person works beyond his or her normal working hours. It can be paid at a rate of time-and-a half, double time, double-time-and-a-half, etc.

Special allowances are often paid to workers. These allowances are additional to their normal pay. Allowances can be paid for dirty work, working in confined spaces, excess heat, working at heights, etc.

e.g.

- 1 A worker has an annual income of \$32 864. Find his holiday pay if he takes 4 weeks holiday and receives $17\frac{1}{2}\%$ loading.

$$\begin{aligned} 4 \text{ week's wage} &= \$ (32\,864 \div 52) \times 4 \\ &= \$2528 \end{aligned}$$

$$\begin{aligned} \text{Holiday loading} &= 17\frac{1}{2}\% \text{ of } \$2\,528 \\ &= \$442.40 \end{aligned}$$

$$\begin{aligned} \text{Holiday pay} &= \$2528 + \$442.40 \\ &= \$2970.40 \end{aligned}$$

- 2 Find the weekly income of a person who is paid \$12.50 an hour for a 38-hour normal week and works 7 hours overtime at time-and-a-half.

$$\begin{aligned} \text{Weekly wage} &= \$ (12.50 \times 38) + \$ (12.50 \times 1\frac{1}{2} \times 7) \\ &= \$606.25 \end{aligned}$$

- 3 A construction worker who works a 38-hour week is paid \$21.50 an hour. In addition, he is paid 38 cents/hour for working on the 17th floor and 48 cents/hour for working in dirty conditions. Determine his weekly wage, if he works his normal 38-hour week, 10 hours of which are on the 17th floor and 15 hours in dirty conditions.

$$\begin{aligned} \text{Wage} &= \$21.50 \times 38 + \$0.38 \times 10 + \$0.48 \times 15 \\ &= \$828 \end{aligned}$$

If leave loading is $17\frac{1}{2}\%$ of 4 weeks pay, calculate the leave loading for a worker on:

- 1 a weekly wage of \$495
- 2 a weekly wage of \$729
- 3 a fortnightly wage of \$1234
- 4 a fortnightly wage of \$1046
- 5 a salary of \$35 568
- 6 a salary of \$48 256.
- 7 Determine the total holiday pay a worker receives if his loading is $17\frac{1}{2}\%$ and his weekly wage is \$795.
- 8 How much is Sharon paid when she leaves on 4 weeks' holiday if her weekly wage is \$1247 and she receives a $17\frac{1}{2}\%$ leave loading?
- 9 Calculate Ita's holiday pay if her leave loading is $17\frac{1}{2}\%$ and her fortnightly wage is \$1452.
- 10 Calculate Martin's holiday pay, if his holiday loading is $17\frac{1}{2}\%$ and his annual salary is \$45 500.

Determine the overtime pay at time-and-a-half for following:

- 11 6 hours at \$14.50 per hour
- 12 7 hours at \$10.50 per hour
- 13 10 hours at \$9.40 per hour
- 14 16 hours at \$13.30 per hour.

Determine the overtime payments for the following, given that the normal rate of pay is \$17.20 an hour:

- 15 3 hours at time-and-a-half and 4 hours at double time
- 16 5 hours at time-and-a-half and 6 hours at double time
- 17 4 hours at time-and-a-half and 3 hours at double time
- 18 6 hours at time-and-a-half, 4 hours at double time and 2 hours at triple time.
- 19 A restaurant worker is paid \$426.90 per week. In addition he receives \$6.47 per broken shift, a meal allowance of \$7.50 per day and a laundry allowance of \$6.90 per week. Determine his pay, if in one week, he works 3 broken shifts and 2 unbroken shifts.
- 20 A sales representative's weekly income is \$484.50. In addition to this he is paid a living away from home allowance of \$36.10 per week, a vehicle allowance of \$186.70 per week plus 18.4 cents per km. Determine his weekly income if he drives 425 km in the week.

The following are examples of allowances that are paid to construction workers:

- Tool allowance – \$16.00 per week
- First Aid Certificate – \$3.19 per day
- Asbestos allowance – \$1.59 per hour
- Wet work – \$0.48 per hour
- Dirty work – \$0.48 per hour.

Determine the weekly wages for four construction workers, who each work a 37½-hour, 5-day week. The table displays their hourly rates along with hours for which they receive allowances:

	Hourly rate	Tool allowance	First Aid Certificate	Asbestos	Wet	Dirty
21	\$16.63	Yes	Yes	10 hours	4 hours	9 hours
22	\$17.20	Yes	No	17 hours	5 hours	4 hours
23	\$18.30	No	Yes	4 hours	10 hours	9 hours
24	\$20.10	Yes	Yes	—	12 hours	14 hours
25	\$21.30	Yes	No	11 hours	6 hours	3 hours
26	\$22.40	No	Yes	15 hours	8 hours	1 hour

- 27 Roger is a construction worker who is paid \$18.20 an hour. He works 38 hours over 5 days and receives a first aid allowance. He spends 5 hours removing asbestos and some hours working in wet conditions. If his weekly pay is \$718.86, how many hours did he work in wet conditions?

Sales representatives are frequently paid a **commission** rather than a set weekly amount. This encourages them to make as many sales as possible.

Authors are paid **royalties**, which are generally a percentage of the total net sales of their books. **Piecework** is paid for each item or unit.

e.g.

- 1 A carpet salesman received a 5.5% commission on his total sales for the week. If his sales totalled \$23 000, how much did he earn in the week?

$$\begin{aligned}\text{Commission} &= \$ (23\,000 \times 0.055) \\ &= \$1265\end{aligned}$$

- 2 Dan is paid 9.5% royalties on the sales of his latest best seller. If the total net sales for the year are \$120 000, determine the royalties Dan receives.

$$\begin{aligned}\text{Royalties} &= \$ (120\,000 \times 0.095) \\ &= \$11\,400\end{aligned}$$

- 3 A fence contractor is paid at the rate of \$4.50 per metre of fencing erected. If in a week he erects 250 metres of fencing, how much does he make in a week?

$$\begin{aligned}\text{Wage} &= \$ (4.50 \times 250) \\ &= \$1125\end{aligned}$$

-
- 1 Hosni is paid a commission of 7.5% of his weekly sales. Determine his weekly income if he sells goods worth \$12 250.
 - 2 Determine the commission paid to a car salesman if the commission is 2.5% and he sells five cars each valued at \$21 200.
 - 3 Find the commission paid on sales of \$2350 at the rate of 12.5%.
 - 4 A financial planner charges an initial fee of 1.8% per annum on investments he makes for his clients. What is the initial fee charged if a client gives him \$320 000 to invest?
 - 5 A financial planner receives a salary of \$42 000 per annum plus he receives 0.55% on investments that he has arranged. How much does he earn in a year if he arranges investments totalling \$5 million.
 - 6 Jenny is paid \$197 a week plus a commission of 13% of her weekly sales. What is her weekly pay if her sales were \$4350?
 - 7 Billie's total income for the week was \$1300. This was made up of \$225 plus 2.5% of his sales. Determine his sales for the week.
 - 8 Gavin's pay for the week was \$1134.50. This consisted of \$187 plus a commission on his sales. Determine his percentage commission if his sales were \$18 950 in the week.
 - 9 Brian sells 2 houses and he receives \$7 888 commission. His commission is 0.8% of his sales. The first house was sold for \$533 500. How much did he receive for the sale of the other house?
 - 10 Shane is paid a commission on his weekly sales. His commission is 10.5% of all his sales. If his commission is \$478.80, determine his weekly sales.

Determine the royalties for each of the following sales:

- 11 14% on sales of \$4500
- 12 13.5% on sales of \$6350
- 13 16.8% on sales of \$1035.
- 14 Amy receives royalties of 9%. Her publisher sells 3500 books for \$25.20 each and sells a further 1250 at \$19.50 each. Determine the total royalties she receives.

- 15 Dan White wrote 4 books. The table below outlines his sales and commission rates. Complete the table and determine his annual average income (to the nearest \$) from royalties, given that the sales are for 3 years.

Book	Sales	Cost of book	Commission rate	Commission
The Leonardo Code	1 340 000	\$14.95	10%	i)
The Numeral Fortress	900 000	\$12.95	9%	ii)
Heavenly Figures	750 000	\$14.50	8.7%	iii)
The Sharp Deceiver	600 000	\$13.95	9.5%	iv)

- 16 Sarah’s books sell for \$40.00 each. She receives 9% royalties. How many books have to be sold so that she receives royalties of \$20 268?
- 17 Boris receives \$3757.25 in royalties on sales of \$39 550. Determine the percentage rate of his royalties.
- 18 A lecturer receives \$1.50 for each examination question he marks. If he marks 35 exam questions per hour in the first 3 hours, 40 questions per hour in the next 4 hours and 43 questions per hour in the next 4 hours, determine his total income and his equivalent hourly rate.
- 19 A barber charges \$12.50 per hair cut and \$2.20 for a beard trim. Determine his income if he does 12 hair cuts and 6 beard trims.
- 20 A fruit picker is paid \$2.50 for each bucket she fills with fruit. Find her weekly income and determine her overall average hourly rate, using the details in the following table:

Day	Number of buckets	Hours worked
Monday	25	7
Tuesday	33	8
Wednesday	28	8
Thursday	35	8
Friday	27	7

- 21 A worker is paid \$3.30 to sew a dress. Determine her income for the week if she makes 215 dresses.
- 22 A timber floor costs \$50/m² for the timber, \$37/m² to lay the floor and \$35/m² to sand and polish the floor. Determine the total cost of buying the flooring and having it laid, sanded and polished if:
- i) the floor has an area of 37 m²
- ii) the dimensions of the room are 7 m by 10.5 m.
- 23 Bob the tiler charges \$50 per m² to lay wall tiles and \$58 per m² to lay floor tiles. If Bob was paid \$1923.40 to tile a bathroom with a floor area of 8.3 m², what is the area of the wall tiles he laid?

Governments pays **allowances** and **pensions** to many eligible people, e.g. youth allowance and old age pension. More information on these and other allowances can be found on the Internet on www.centrelink.gov.au. For this topic you will need to refer to **tables 1, 2, 3 and 4** on pages 38–39.

e.g. Dave is a 19-year-old student and lives at home.

- i) How much is his youth allowance if he has no other income? Answer: \$220.30
- ii) How much is he able to earn before losing his entire allowance? Answer: \$573.57
- iii) What is the maximum he can earn and still keep the full youth allowance? Answer: \$236
- iv) If Dave earns \$370 per fortnight, calculate his new youth allowance.

$$\begin{aligned} \text{Reduction above \$316} &= (\$370 - \$316) \times 0.70 \\ &= \$37.80 \end{aligned}$$

$$\begin{aligned} \text{Reduction between \$236 and \$316} &= (\$316 - \$236) \times 0.50 \\ &= \$40 \end{aligned}$$

Dave's allowance is reduced by \$77.80

$$\begin{aligned} \text{Dave's new allowance} &= \$220.30 - \$77.80 \\ &= \$142.50 \end{aligned}$$

For each of the following, state:

- i) The youth allowance if the person has no other income
 - ii) The allowable income to still receive the full youth allowance
 - iii) The income that will make a person ineligible for the youth allowance.
- 1 A 17-year-old student living at home.
 - 2 A 20-year-old job seeker living away from home.
 - 3 A single student with one child.
 - 4 A student with a partner and one child.
 - 5 Sally is 18 years old and is eligible for the youth allowance. If she is a job seeker and living at home and she earns \$53 per fortnight, how much is her youth allowance for the fortnight?
 - 6 Anwar is 18, a university student and living at home. He earns \$150 in a fortnight at his casual job. What is his fortnightly youth allowance?
 - 7 Peta is 17, single and a job seeker. If she is living with her parents and earns \$130 per fortnight doing casual work, by how much will her fortnightly youth allowance be reduced?
 - 8 George is 19, and receives a youth allowance. He is a student and lives at home. He earns \$160 in a fortnight. What will be the reduction in his youth allowance per fortnight?
 - 9 Sarah is 19 and lives at home. She attends a T.A.F.E. College. What is her youth allowance if she works at the local corner store and earns \$300 a fortnight?
 - 10 Dawn is a job seeker and has a 2-year-old child. She is eligible for a youth allowance and lives with her parents. She earns \$140 a fortnight doing casual work. What is her fortnightly allowance?
 - 11 Connor is 19, a job seeker and living away from home. What is his youth allowance if he earns \$230 working casually at the nearby fruit barn?
 - 12 Mandy receives a single old age pension. What is her pension if she earns \$100 per fortnight?
 - 13 Fred is a single old age pensioner. He has 1 dependent child. Find his pension if he earns \$180.60 per fortnight.
 - 14 A married couple receive the old age pension. What is their pension per fortnight if:
 - i) they have 1 dependant child and earn \$240 per fortnight
 - ii) they have 1 dependent child and earn \$280.60 per fortnight
 - iii) they have 3 dependent children and earn \$1000.80 per fortnight.
 - 15 How much can a couple with 3 dependent children earn:
 - i) and still receive the full old age pension?
 - ii) to be ineligible to receive the aged pension?

Gross pay is your income before deductions. **Net** income is gross income minus deductions. Deductions may include income tax, union fees, superannuation, medical benefit contributions, HECS fees, etc.

e.g. Joanne's gross fortnightly wage is \$1520. Her deductions per fortnight are \$277.15 tax, \$30.50 superannuation, \$8.50 union fees and medical insurance of \$19.50. Find her net income.

$$\begin{aligned}\text{Net pay} &= \text{Gross pay} - \text{deductions} \\ &= \$1520 - (\$277.15 + \$30.50 + \$8.50 + \$19.50) \\ &= \$1184.35\end{aligned}$$

- Dave's gross pay is \$760 per week. Determine his net weekly pay if he pays \$138.58 tax, \$45 in superannuation, \$8.70, union fees and \$23 medical insurance.
- Anne has a gross weekly pay of \$840. She pays \$35 in superannuation, \$162.58 tax and \$10.21 in union fees. Calculate her net weekly wage.
- Robin has a fortnightly gross wage of \$1250. His deductions include union fees of \$8.90, superannuation of \$38.50, medical of \$11.34 and tax of \$196.15. What is his net pay?
- Freddie's gross income per fortnight is \$1300. He pays superannuation of \$34.50, medical insurance of \$13.40 and tax of \$211.15 each fortnight. What is his net fortnightly income?
- Hayley's weekly pay is \$980. She pays \$204.58 in tax, \$71.30 in medical insurance, \$9.50 in union fees, \$23.50 in superannuation and 6.5% HECS payment. What is her net weekly pay?
- Shane pays 4% of his gross fortnightly pay on his HECS debt. He also pays \$12.30 medical insurance, \$14.50 union fees and tax of \$235.15. Determine his net fortnightly pay if his gross fortnightly pay is \$1380.
- Harrison's net pay is \$586 per week. His weekly deductions are \$153.55 tax, \$41.50 superannuation, \$12.50 union fees and \$16.50 in medical insurance. Determine his gross weekly pay.
- Keith's net pay per week is \$812. His deductions per week are tax, \$239.72, union fees, \$8.75 and superannuation, \$37.00. Calculate his gross weekly income.
- Kim pays 4% of her gross pay on superannuation. Her gross weekly income is \$1250. Her other weekly deductions include \$15.40 on medical insurance, \$15.60 on union fees and \$285.58 in taxation. Calculate her net weekly income.
- Yvette has a salary of \$45 000. Her tax bill is \$8850, medical insurance is \$899.60, 5% of her salary goes to her HECS debt, and 5% goes in superannuation. Determine her annual net pay and her monthly net income.
- Joey has a weekly gross of \$1345. His superannuation contribution is 5% of his gross income. He has deductions of \$20.90 for medical insurance, union fees of \$15.20 and taxation of \$314.08. Calculate his net weekly income.

Complete the following table to find the net weekly incomes:

Name	Gross income	Union fees	Super	HECS	Medical insurance	Tax	Net income
12 Annie	\$498	\$5.40	\$24	Nil	Nil	\$59.98	
13 Brian	\$674	\$7.40	\$33	4%	Nil	\$112.28	
14 Caitlyn	\$875	\$9.20	Nil	5.5%	\$5.83	\$173.08	
15 Danny	\$973	\$9.20	\$61	6%	\$6.88	\$202.48	
16 Emma	\$1145	\$10.20	\$48	Nil	\$11.30	\$254.08	
17 Francis	\$1250	\$7.50	\$73	Nil	\$9.60	\$285.58	
18 Gemma	\$1380	Nil	Nil	8%	\$13.50	\$324.58	
19 Harry	\$1760	Nil	\$81	Nil	\$13.50	\$470.35	

- 1 Lance is paid \$16.50 an hour at the local restaurant. If he works a 35-hour week, determine his weekly wage.
- 2 Ivan is paid \$14.50 an hour at a sandwich shop. He works a $37\frac{1}{2}$ -hour week. Calculate his weekly wage.
- 3 Shelby earns \$14.10 an hour. She needs to earn \$556.95. How many hours must she work?
- 4 Nana works from 7.30 am to 4.00 pm, Monday to Friday. She has an unpaid lunch break of 45 minutes. Her hourly rate of pay is \$19.60. Calculate:
 - i) her daily wage
 - ii) her weekly income
 - iii) her annual income.
- 5 Hiam earns an annual salary of \$37 869. Determine his fortnightly pay.
- 6 Complete Spike's weekly time sheet and determine his annual wage if he works the same times each week and is paid an hourly rate of \$14.86.

Day	Start	Finish	Hours worked
Monday	8.00 am	4.00 pm	i)
Tuesday	7.00 am	3.00 pm	ii)
Wednesday	7.00 am	3.15 pm	iii)
Thursday	8.00 am	4.30 pm	iv)
Friday	7.00 am	2.30 pm	v)
TOTAL			vi)

- 7 Charlotte's weekly income is \$485.50.
 - i) What is her annual salary?
 - ii) She agrees to have her income paid monthly. How much will she receive a month?
- 8 Chris receives his salary monthly. At present he receives \$4394 a month. His employer decides to pay him fortnightly. How much would Chris receive a fortnight?
- 9 Winston receives a gross weekly wage of \$980. He receives a pay rise that increases his annual salary to \$53 253.20. What has his percentage wage increase?
- 10 Ahmed has a choice of wages. Which is the greater annual wage and by how much?
 - i) \$14.95 an hour for a 38-hour week (assuming he is paid for 52 weeks)
 - ii) A monthly wage of \$2529
- 11 Aninda's gross income is \$860 per week. From this he pays \$168.58 in tax, union fees of \$8.50, superannuation of \$43.50 and medical insurance of \$7.50. Determine his net weekly income.
- 12 Anna's fortnightly deductions from her pay are \$54 in superannuation, \$12.50 for medical insurance, \$8.50 for union fees and \$107.88 income tax. If her gross pay is \$950, calculate her net fortnightly pay.
- 13 Delta pays 4% of her gross salary in superannuation. Her other weekly deductions are union fees of \$8.70, medical insurance of \$16.50 and tax of \$330.58. If her gross pay is \$1400, determine her net weekly pay.
- 14 Nassar receives a net weekly wage of \$504.32. His tax was \$104.14. He also paid \$12.50 p.w. for medical insurance and \$24.25 p.w. for superannuation. Determine his gross annual salary.
- 15 Sue is paid \$2259.50 per fortnight and Barry is paid \$4732.90 per month. Who has the higher income and by how much?

- 16 Sherlock receives a $17\frac{1}{2}\%$ leave loading for his 4 weeks annual holidays. His annual salary is \$40 872. Determine the total gross pay he receives when he takes his annual holidays.
- 17 Indigo is paid overtime for working on Saturdays. She is paid time-and-a-half for the first 4 hours, then double time for the rest of the time. Her weekly working hours are shown in the table below. What is her weekly income if she is paid \$11.50 per hour?

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
No. of hours	$7\frac{1}{2}$	8	4	6	8	6

- 18 Sarah works a 37-hour week and her hourly rate is \$16.50. If she works more than 37 hours, she is paid overtime at the rate of time-and-a-half for the first 4 hours and double time for each hour after that.
- Find her normal weekly wage.
 - Find her wage if she works 40 hours in a week.
 - Find her wage if she works 46 hours in a week.
 - If she was paid \$795.60 in one week, how many hours did she work?
- 19 Toby is a construction worker. He is paid the following allowances:
- a tool allowance of \$15.50 per week
 - \$1.59 an hour for working with asbestos
 - \$0.53 an hour for working in wet conditions.
- His normal pay rate is \$16.50 per hour for a $37\frac{1}{2}$ -hour week. Determine his weekly wage if he works 10 hours with asbestos and 5 hours in wet conditions.
- 20 Mahatir is a sales representative. His base wage is \$560 per week. He is paid a living away allowance of \$46.50 per week; a vehicle allowance of \$158.60 a week plus 19.5 cents/km. Determine his pay for the week where he travels 586 km.
- 21 Jason is a car salesman. He receives a commission of 2.5% on each car he sells. If in one week he sells cars for \$17 400, \$21 555 and \$37 485, determine his commission.
- 22 A bricklayer receives \$95 per 100 bricks laid. A house requires 4750 bricks. How much does the bricklayer receive for the house?
- 23 An author is paid 15% royalties on the sales of his books. If his publisher sells 2150 books at \$15.50 each, and 4150 books at \$18.30 each, determine his royalties.
- 24 A tiler is paid \$31.50 per square metre to lay tiles. He lays tiles on a bathroom floor that is 6 m by 3 m and a veranda 17.6 m by 2.4 m. How much does he receive in total?
- 25 George is a 17-year-old student living at home with his parents.
- How much per fortnight is his youth allowance if he earns \$180 per fortnight?
 - How much would his youth allowance be if he turns 18, is a job seeker and still earns \$180 per week?
- 26 Jayden is single, living at home and a new apprentice. Determine his youth allowance if:
- he is 17 and earning \$522 per fortnight
 - he is 18 and earning \$522 per fortnight.

For the following questions, refer to **Tables 1, 2, 3 and 4** on pages 38–39.

- 27 Jane receives a youth allowance. She is single and has 2 children. What is her fortnightly allowance?
- 28 Freda is a 19-year-old student living at home. She receives a youth allowance. She earns \$410 a week from her part-time job. Calculate her youth allowance.
- 29 David is an old age pensioner. How much pension does he receive if he earns \$145 a week mowing lawns?
- 30 Thelma and Tom are married with two children. They receive an old age pension. Determine the pension they receive if they earn a combined income of \$340 a week.

Refer to the **water rates account** on page 40.

- 1 How can payment be made by credit card?
- 2 What percentage of the bill was the fixed charge?
- 3 How many kilolitres were used in this billing period?
- 4 How many days were in this billing period?
- 5 If 85 kilolitres were used, calculate the daily average usage.
- 6 What is the water usage rate?
- 7 What is the sewerage usage rate?
- 8 What percentage of the water used is the discharge factor for the sewer usage?
- 9 If 96 kilolitres of water were used in the billing period, calculate the total usage charge, for both water and sewerage usage.
- 10 How many more litres of water were used in the last billing period, assuming the number of days in the two billing periods was the same?

Refer to the **council rates & charges notice** on page 41.

- 11 What is the account number?
- 12 When is the first instalment due?
- 13 How much is the first instalment?
- 14 How many instalments are required to pay the total bill?
- 15 Does it cost more to pay the bill by instalments, compared to paying the total amount?
- 16 Approximately how much time is between instalment payments?
- 17 What components of the rates are based on the land value?
- 18 What is the value of the land?
- 19 What is the interest rate on overdue rates, and how often is it calculated?
- 20 If the value of the land was \$154 000, calculate the amount of the Ad Valorem rate and the Environmental levy.

Refer to the **quarterly electricity tax invoice** on page 42.

- 21 What is the total amount payable on this bill?
- 22 What is the account number?
- 23 Was the last bill paid in full?
- 24 When is this account due?
- 25 What is the average daily consumption for this bill?
- 26 How many kilograms of greenhouse gases were emitted to provide the energy used in this period?
- 27 If a discount had not been given, what would be the total amount payable?
- 28 Calculate the percentage decrease in average daily usage of this bill, when compared to the same period last year.

A **budget** helps to keep a person’s income and expenditure under control, so that they can manage their money. This way financial difficulties such as credit card debt can be avoided.

- 1 Carol has a net weekly income of \$690. She has some shares and money invested that generate \$550 per annum. Complete the budget for Carol’s yearly income.

Annual Income = i) \$..... + ii) \$..... = ...\$36 430

		Annual expenditure
Rent	\$150 per week	iii) \$.....
Water	\$100 per quarter	\$400
Electricity	\$280 per quarter	iv) \$.....
Telephone	\$55 per month	v) \$.....
Insurance – home	\$250 per year	\$250
– contents	\$170 per year	\$170
– medical	\$11.20 per week	vi) \$.....
– car	\$520 per year	vii) \$.....
Petrol and car repairs	\$30 per week	viii) \$.....
Car registration	\$450 per year	\$450
Food	\$120 per week	\$6240
Clothes/footwear	\$60 per week	\$3120
Sport/equipment	\$20 per week	ix) \$.....
Entertainment	\$40 per week	\$2080
Hair cuts/make-up	\$10 per week	\$520
Miscellaneous	\$50 per week	\$2600
	TOTAL =	\$29 112.40

Carol wants to go overseas sometime in the future. She estimates she will need about \$22 000 for the planned trip. If her expenditure and income remain the same, approximately how many years will it take Carol to save the required money?

For each of the following scenarios, draw up a budget table. Answer the questions associated with each scenario.

- 2 Will’s Income: net fortnightly – \$1760
interest on investments - \$750 p.a.

Expenditure: house repayments – \$1600 per month
water – \$180 per quarter
gas – \$110 per quarter
house insurance – \$870 p.a.
car repairs and petrol – \$90 per week
food – \$125 per week
entertainment – \$50 per week

land rates – \$240 per quarter
electricity – \$185 per quarter
telephone – \$120 per month
car insurance – \$850 p.a.
car registration – \$560 p.a.
clothes/shoes – \$78 per week
medical – \$30 per month

How much does Will expect to save in the year?

- 3** Zoe's Income: net weekly \$684
interest on investments – \$321 p.a.

Expenditure:	rent – \$137 per week	water – \$89 per quarter
	electricity – \$120 per quarter	telephone – \$35 per week
	contents insurance – \$310 p.a.	car expenses – \$80 per week
	car repayments – \$56 per week	petrol – \$50 per fortnight
	food – \$80 per week	gym – \$180 per quarter
	clothes – \$75 per week	entertainment – \$50 per week
	haircuts/make-up – \$80 per month	miscellaneous – \$40 per week

If her rent increases by \$25 per week and Zoe purchases a new car that will take her repayments up to \$63 per week, how much will her savings be reduced to in the full year, assuming her income and expenditure remain the same?

- 4** Don's Income: \$701 net per week
interest on investments – \$484 p.a.

Expenditure:	mortgage repayments – \$120 per week	water – \$140 per quarter
	electricity – \$195 per quarter	gas – \$80 per quarter
	mobile phone – \$55 per month	insurances – \$1200 p.a.
	medical – \$12 per week	car repairs – \$50 per week
	car repayments – \$120 per week	registration/licence – \$525 p.a.
	food – \$100 per week	clothes/shoes – \$45 per week
	entertainment – \$80 per week	incidentals – \$35 per week

Don needs to save \$6552 for a holiday in 2 years. How much a week must he save? Is this weekly saving possible if his income and expenditure do not change in this period? Justify your answer.

- 5** Mandy's Income: \$800 net per week
interest on investments: \$140 p.a.

Expenditure:	rent – \$180 per week	water – \$150 per quarter
	electricity – \$120 per quarter	gas – \$78 per quarter
	telephone – \$71 per month	contents insurance – \$210 p.a.
	medical – \$40 per week	car insurance – \$910 p.a.
	car repairs/petrol – \$90 per week	car repayments – \$220 per month
	car registration – \$550 p.a.	food – \$85 per week
	clothes/footwear – \$80 per week	sport/equipment – \$15 per week
	entertainment – \$90 per week	miscellaneous – \$100 per week

Is Mandy able to save anything if she stays with this budget?

- 6** Gautum's Income: \$1400 net per fortnight
interest on investments: \$250 p.a.

Expenditure:	rent – \$250 per week	water – \$163 per quarter
	electricity – \$174 per quarter	gas – \$71 per quarter
	telephone – \$75.50 per month	contents insurance – \$52.60 per month
	car insurance – \$810 p.a.	medical insurance – \$35.50 per week
	car expenses – \$46 per week	car repayments – \$110 per month
	car registration – \$580 p.a.	donations to charity – \$15 per week
	food – \$76 per week	entertainment – \$81 per week

- Calculate Gautum's savings per week.
- Gautum decides to buy an investment property. His repayments on the loan would be \$250 per week, and land rates and insurance \$1500 per year. How much rent would he need to charge per week (to the nearest dollar) in order to pay off the loan and maintain the level of his savings per week?

Simple Interest formula: $I = Prn$, where I = interest P = Principal
 r = interest rate per period as a decimal
 n = number of periods

or $I = \frac{Prn}{100}$, where r = interest rate

e.g.
1 Calculate the simple interest on \$4500 if it is invested for 3 years at a rate of 8.75% p.a.

$$I = \$ (4500 \times 0.0875 \times 3) \quad \text{or alternatively, } I = \frac{\$4500 \times 8.75 \times 3}{100}$$

$$= \$1181.25 \quad = \$1181.25$$

2 Calculate the simple interest on \$7500 if it is invested for 7 months at a rate of 7.8% p.a.

$$I = \$7500 \times 0.078 \times (7/12) \quad \text{or alternatively, } I = \frac{\$7500 \times (7.8 \div 12) \times 7}{100}$$

$$= \$341.25 \quad = \$341.25$$

Calculate the simple interest if:

- 1 \$4500 is invested at 6.5% p.a. for 3 years
- 2 \$7500 is invested at 7.6% p.a. for 5 years
- 3 \$950 is invested at 10.1% p.a. for 3 years
- 4 \$1256 is invested at 9.8% p.a. for 10 years
- 5 Calculate the simple interest if \$7560 is invested at a rate of 8.25% p.a. for a period of 9 years.
- 6 Determine the simple interest if \$8450 is invested for 3 years at an interest rate of 4.7% p.a.
- 7 Calculate the simple interest on \$1450 invested for 6 years at 3.75% p.a.
- 8 If Joe invested \$7640 at a simple interest rate of 7.6% p.a., how much would he have after 10 years?
- 9 Jane invested an amount of money at 5% p.a. After 7 years she had received \$1673 in interest. How much did she invest?
- 10 Danny invested \$7500 in an account for a period of 6 years. If after that time he received \$4275 in interest, calculate the simple interest rate paid on the account.
- 11 Lindy invested \$7846 in an account that paid simple interest at a rate of 8.5% p.a. For how many years was the money invested if the investment earned \$6002.19 in interest?
- 12 Complete the table below by filling in the related interest rates:

Annual	Monthly	Half-yearly	Quarterly
10.8%			
	1.2%		
		4.5%	
			4.035%

- 13 If \$1750 was invested for 6 months at a simple interest rate of 15% p.a., how much interest would be earned?
- 14 Determine the simple interest when \$8360 is invested at 10.5% p.a. for:
 - i) 9 months ii) 11 months.
- 15 If Judy invests \$1740 at a simple interest rate of 7.2% p.a. how much would she have in total after:
 - i) 7 months ii) 15 months iii) 2 years and 2 months?
- 16 Sara invests money in an account that pays simple interest. If she invests \$4860 at 6.0% p.a., how many months must she leave her money there to receive \$121.50 in interest?
- 17 Chris invests \$7800 for 15 months and receives \$877.50 in interest. Determine the simple interest rate per annum.
- 18 Myra invests \$9760 in an account paying simple interest at a rate of 12% p.a. If she receives \$1073.60 in interest, determine the number of months she had her money invested.

Compound interest is sometimes referred to as interest earned on interest. It may be calculated year by year, or by the use of a formula.

e.g. Determine the interest earned when \$1500 is invested at 6% p.a. compound interest for 3 years.

Year by year:

Principal beginning of Year 1	=\$1500
Interest at end of Year 1	=\$90 (1500×0.06)
Principal beginning of Year 2	=\$1590
Interest at end of Year 2	=\$95.40 (1590×0.06)
Principal beginning of Year 3	=\$1685.40
Interest at end of Year 3	=\$101.12 (1685.40×0.06)
Amount at end of 3 years	=\$1786.52
Interest earned in 3 years	=\$1786.52 - \$1500
	=\$286.52

Formula: $A = P(1 + r)^n$, where

A = final amount (future value)

P = the principal

n = number of compounded periods

r = interest rate as a decimal

$$A = \$1500 \times (1.06)^3$$

$$= \$1786.52$$

$$\text{Interest} = (1786.52 - 1500)$$

$$= \$286.52$$

- Find the compound interest on \$1400 invested for 2 years at 10% p.a. Record the interest earned year by year in the table below.

Principal beginning of Year 1	
Interest at end of Year 1	
Principal beginning of Year 2	
Interest at end of Year 2	
Principal at end of Year 2	

- Find the compound interest on \$2500 invested for 3 years at 9% p.a. (Use the year-by-year method.)
- Determine the interest earned if \$7860 is invested for 5 years, compounding yearly at 9% p.a.
- If \$8970 is invested for 6 years, compounding yearly, at an interest rate of 5% p.a., calculate the interest.
- Calculate the amount \$6500 will grow to in 5 years, if it is invested at 9.5% p.a. compounding annually.
- Determine the amount \$7800 will grow to if it is compounded at 6% p.a. for
 - 3 years
 - 10 years
 - 35 years
- \$9350 is invested at 9% p.a. for 12 years. Calculate the interest earned if i) the interest is paid as simple interest and ii) the interest is compounded yearly. What is the difference between these two amounts?
- If interest is compounded quarterly, determine the interest earned when \$7680 is invested at 8% p.a. for 5 years (Hint: Find the equivalent interest rate per quarter, and find how many quarters there are in 5 years.)
- \$9370 is invested for 6 years at 9% p.a. Calculate the interest earned if interest is compounded each 6 months.
- Calculate the amount \$4850 will grow to, invested at 9% p.a. for 6 years with interest compounded monthly.
- Calculate the interest when \$8580 is invested for 10 years at 8% p.a., with interest compounded quarterly.
- Calculate the interest when \$7500 is invested for 12 years at 8% p.a., if interest is compounded monthly.
- If interest is compounded every 6 months, what will \$6450 amount to in 3 years at 8.15% p.a.?
- David invests \$9000 for 10 years at an interest rate of 7% p.a., compounding annually. Pam invests the same amount for the same period and interest rate, but her investment is compounding monthly. Determine who earns the greater amount of interest and what is the difference?

Income tax is calculated according to the following table:

Income Range (\$)	Tax
0–6000	Nil
6001–25 000	15¢ for each \$1 over \$6000
25 001–75 000	\$2850 plus 30¢ for each \$1 over \$25 000
75 001–150 000	\$17 850 plus 40¢ for each \$1 over \$75 000
Over 150 000	\$47 850 plus 45¢ for each \$1 over \$150 000

e.g.

1 Use the tax table to calculate the tax payable on

- i) \$5400 As in the \$(0–6000) tax bracket, no tax payable
- ii) \$52 000 \$52 000 is in the \$(25 001–75 000) tax bracket
 $\text{Tax} = \$2850 + \$[(52\,000 - 25\,000) \times 0.30]$
 $= \$10\,950$
- iii) \$81 550 \$81 550 is in the \$(75 001–150 000) tax bracket
 $\text{Tax} = \$17\,850 + \$[(81\,550 - 75\,000) \times 0.40]$
 $= \$20\,470$

The **Medicare levy** is payable by all taxpayers in Australia whose taxable income is more than \$16 284. For those whose taxable income is greater than \$16 284 but less than or equal to \$17 604, the levy is 20 cents in the dollar for every dollar in excess of \$16 284. For incomes greater than \$17 604 the levy is 1.5% of the taxable income.

2 Calculate the Medicare levy if the taxable income is:

- i) \$15 600 Since \$15 600 is below the threshold, no levy is payable
- ii) \$16 450 $\text{Levy} = \$ (16\,450 - 16\,284) \times 0.20 = \33.20
- iii) \$28 450 $\text{Levy} = 1.5\% \text{ of } \$28\,450 = \$2\,845 \times 0.015 = \426.75

Determine the income tax payable on each of these amounts.

- 1** \$4750 **2** \$13 850 **3** \$63 040 **4** \$95 420 **5** \$159 400

6 Danny earns \$650 a week. **i)** Calculate his annual salary. **ii)** Calculate his yearly tax payable.
iii) Calculate his weekly take-home pay after tax has been deducted.

7 Fred earns \$980 gross per week. Freda earns \$54 450 a year. Determine the difference in their weekly take-home pay after tax has been deducted.

8 Calculate the tax payable each week on a salary that pays \$1056 per week.

Determine the Medicare levy on each of these taxable incomes.

- 9** \$5850 **10** \$16 284 **11** \$17 604 **12** \$17 912 **13** \$21 700 **14** \$73 250

Determine the net annual salary, after tax and the Medicare levy, for an employee who earns:

15 a yearly salary of \$21 550 **16** a yearly salary of \$36 550

17 a yearly salary of \$75 650 **18** a fortnightly salary of \$2156

19 Dave's fortnightly salary is \$2714. Determine his net fortnightly pay after tax and Medicare levy have been deducted.

20 Lance receives a salary of \$1475 a fortnight. If during the year he also receives \$595 in interest on his investments, calculate: **i)** his total annual income
ii) the tax payable on this income, assuming he has no deductions
iii) his Medicare levy **iv)** his total net yearly income.

21 If Rochelle's weekly wage before tax is \$758, determine her annual income after tax and the Medicare levy. Hence determine her net weekly pay.

22 Mo's fortnightly salary is \$2884. Find his fortnightly salary after tax and Medicare levy have been deducted.

(To complete this topic, you need the Tax Table and Medicare Levy data from topic 12.)

Work related expenses reduce taxable income. These expenses include donations over \$2.00 to registered charities, car expenses, protective clothing, laundry, sun protection, self-education expenses, reference books, union fees, computers, attendance at seminars, tax accountants fees, journals etc.

e.g. John has a gross income of \$45 750. He also receives \$750 in interest from his investments. His employer deducts \$9700 in tax, and he makes the following claims as tax deductions: car expenses of \$250, laundry expenses of \$150, union fees of \$380, self-education expenses of \$180 and tax accountant fees of \$102.

Calculate:

- i) his taxable income $\text{Taxable Income} = \text{Income} + \text{Interest} - \text{Deductions}$
 $= \$ (45\,750 + 750) - \$ (250 + 150 + 380 + 180 + 102) = \$45\,438$
- ii) the income tax he must pay on this income
 $\text{Tax payable} = \$2850 + \$ (45\,438 - 25\,000) \times 0.30 = \8981.40
- iii) Medicare levy $\text{Medicare levy} = 1.5\% \text{ of } \$45\,438 = \$681.57$
- iv) the amount of additional tax or tax refund due.
 $\text{Total tax payable} = \$ (8981.40 + 681.57) = \9662.97

As this is less than the tax he has already paid, he will receive a refund of \$37.03

For questions 1–5, find

- i) the taxable income
 - ii) the tax payable on the taxable income
 - iii) the Medicare levy
 - iv) tax refund or additional tax payable.
- 1 Annual salary of \$14 500 and income from investments \$185. Deductions of union fees \$125 and travel expenses \$180. Tax paid \$1 290.
 - 2 Annual salary of \$16 800 and interest received \$130. Deductions of union fees \$120, tool allowance \$140, protective clothing \$250 and laundry \$98. Tax already paid \$1620.
 - 3 Annual salary of \$26 500 and interest from bank account \$280. Tax paid \$3300. Deductions are union fees of \$220, sun protection \$56 and journals and newspapers, \$150.
 - 4 Annual salary of \$56 750. Tax paid \$13 300. Interest received, \$90. Deductions are travel \$180, computer depreciation of \$350 and self-education expenses of \$380.
 - 5 Annual salary \$75 300 and extra income \$1450. Deductions for self-education \$380, newspapers \$150, tax agent \$130 and attendance at a work related conference \$480. Tax instalments deducted totalled \$19 100.
 - 6 Daniel has an income of \$48 370. He also had a second job which paid him an extra \$3480. He has allowable deductions of \$984 and his tax instalments total \$12 486. Calculate Daniel's tax refund.
 - 7 Samantha has full-time employment which pays her \$78 354 a year. She has additional income from investments and royalties of \$15 450, for which she has paid no income tax. Samantha's employer had deducted \$20 400 in tax instalments. She has the following allowable tax deductions – travel \$680; reference books \$460; attendance at seminars \$750; protective clothing \$490; laundry \$140; tax agent's fees \$215. Calculate how much Samantha has to pay the taxation department.
 - 8 Sahir has an income of \$157 400. He has additional income of \$1470 from interest on his investments. His allowable deductions are union fees of \$270; journals and reference books totalling \$480; seminar attendance \$480 and tax accountant's fees of \$190. If his employer has deducted \$53 540 in tax instalments, calculate any refund Sahir may receive or the additional tax he must pay.

- 1 Raj has a net annual income of \$35 619. He also has annual interest from an investment of \$285. He wishes to set up a budget for the coming year based on his expenditure for this year. His expenses are as follows:

Rent – \$180 per week	water – \$120 per quarter
electricity – \$140 per quarter	gas – \$98 per quarter
telephone – \$25 per week	contents insurance – \$160 p.a.
car repairs/petrol – \$80 per week	car insurance – \$49 per month
car repayments – \$120 per week	registration and licence – \$580 per year
food – \$68 per week	clothes/footwear – \$25 per week
sports/equipment – \$15 per week	entertainment – \$90 per week
medical – \$30 per week.	

Complete a budget for the year. Raj wishes to save \$800 in the year. He decides to reduce his entertainment expenses to do his. By how much per week, must he reduce his expenditure on entertainment in order to reach his goal?

- 2 Refer to the **aussie telco account** on page 43.
- Was the previous bill paid in full?
 - What is the difference between this bill and the last bill?
 - By how much did the Internet usage increase?
 - How much GST was paid?
 - What is the approximate difference in the bill total between last June and this June?
 - Which phone number has the greatest decrease in usage?
 - If 16 calls were made from the mobile phone, what was the approximate average cost per call?
- 3 Calculate the simple interest of \$4500 if it is invested for 10 years at 12% p.a.
- 4 If Steve invests \$10 750 for 6 months at a simple interest rate of 18% p.a., how much interest will he receive?
- 5 Jenny invests \$17 200 at a simple interest rate of 14% p.a. for 14 months. How much interest will she receive?
- 6 Bryson invests \$14 000 at a simple interest rate of 7.5% p.a. How much will this amount to after 2½ years?
- 7 Calculate the interest on \$6890 if it is invested at 9.2% p.a. compounding annually for 6 years.
- 8 Jacinta invests \$7580 at 6.75% p.a. If the interest is compounded annually, how much will Jacinta have after 7 years?
- 9 Fred invests \$20 000 for 5 years at 8% p.a. If the interest compounds every 6 months, calculate the interest earned.
- 10 Nicola needs to have \$12 000 at the end of 4 years. How much money must she invest at 12% p.a. if the interest is compounded annually?

For the following questions, you may need to use the tax table below and the Medicare details on the next page.

Income Range (\$)	Tax
0–6000	Nil
6001–25 000	15¢ for each \$1 over \$6000
25 001–75 000	\$2850 plus 30¢ for each \$1 over \$25 000
75 001–150 000	\$17 850 plus 40¢ for each \$1 over \$75 000
Over 150 000	\$47 850 plus 45¢ for each \$1 over \$150 000

Medicare levy

- For those on incomes up to and including \$16 284 there is no Medicare levy.
- For those whose income is greater than \$16 284 but less than or equal to \$17 604, the levy is 20 cents in the dollar for every dollar in excess of \$16 284.
- For incomes greater than \$17 604 the levy is 1.5% of the taxable income

- 11** Financial advisers often talk about the 'magic of compounding', and encourage people to start saving for their retirement at the earliest possible age. David and Diane both plan to retire at age 60. They both invest \$3500 at 8.9% p.a. compounding annually.

David invests his money on his 30th birthday, and Diane invests her money on her 18th birthday.

How much more than David will Diane collect (to the nearest \$) when she retires on her 60th birthday?

Calculate the tax payable on these incomes.

- 12** \$5761
13 \$17 480
14 \$38 950

Calculate the Medicare levy on these taxable incomes.

- 15** \$14 900
16 \$16 780
17 \$47 500
18 Calculate the net weekly income after tax and Medicare levy have been deducted on the following:
- i) a gross weekly wage of \$1096
 - ii) a gross fortnightly wage of \$3230
 - iii) a monthly wage of \$6250
 - iv) an annual salary of \$16 328

- 19** If Joe earns a gross income of \$36 750, determine

- i) his tax payable.
- ii) his Medicare levy.

- 20** A taxpayer earns \$48 760 and has the following deductions – union fees \$380, journal subscription \$168, car expenses \$372, protective clothing and laundry \$489 and sun protection \$48.

Calculate:

- i) his taxable income.
- ii) his tax payable.
- iii) his net income, after paying his tax and his Medicare levy.

- 21** Sarah has an income of \$37 638. She has additional income of \$2 380 from other sources. She has the following deductions—protective clothing \$380, reference books \$195, travel expenses \$327, union fees \$280. Her employer has deducted \$7 715.63 in tax and Medicare instalments. Will Sarah receive a tax refund or will she have to pay additional tax? Calculate how much this might be.

Governments have many ways of raising revenue apart from income taxes. A **goods and services tax (GST)** is also paid in Australia. In many countries, a **value added tax (VAT)** is a similar tax. In Australia the GST is 10%. The VAT for some other countries is as follows:

Austria–20%	Denmark–25%	France–19.6%	Spain–16%
U.K.–17.5%	Poland–22%	Estonia–18%	Norway–24%

In Australia, the GST is included in the advertised price. In many other countries the VAT is added on after the advertised price.

e.g.

1 The cost of a television before GST is \$589. How much would the advertised price be, after the GST is added?

$$\begin{aligned}\text{Price} &= \$589 + 10\% \text{ of } \$589 \\ &= \$589 + 0.10 \times \$589 \\ &= \$589 \times 1.1 \\ &= \$647.90\end{aligned}$$

2 If the same television were bought in Poland, what would be the price after VAT is added?

$$\begin{aligned}\text{Price} &= \$589 + 0.22 \times \$589 \\ &= \$589 \times 1.22 \\ &= \$718.58\end{aligned}$$

3 Joe bought a television in England. He paid \$763.75. How much VAT did he pay?

$$\begin{aligned}\$763.75 &= \text{Price} + 17.5\% \text{ of Price} \\ &= \text{Price} \times (1 + 0.175)\end{aligned}$$

$$\begin{aligned}\text{Thus Price} &= \$763.75 \div 1.175 \\ &= \$650\end{aligned}$$

$$\begin{aligned}\text{VAT paid was } & \$763.75 - \$650 \\ &= \$113.75\end{aligned}$$

NOTE: If the GST is 10%, dividing the advertised price by 11 will give the value of the GST.

Determine the GST which must be added by the manufacturer to these items.

- | | |
|-----------------------------------|---|
| 1 a TV valued at \$890 | 2 a \$1.50 bus ticket |
| 3 a car worth \$25 850 | 4 a \$1.00 newspaper |
| 5 a bicycle valued at \$56 | 6 a barbeque chicken costing \$4.50. |
- 7** Calculate the VAT amounts payable on a bike, if the cost of the bike is \$A420, and it were purchased in each of the following countries:
- | | |
|---------------------|-------------------|
| i) Austria | ii) Norway |
| iii) Estonia | iv) UK |
| v) France | vi) Spain. |

Calculate the GST paid on each of these items.

- | | |
|---|---|
| 8 a bus ticket costing \$2.20 | 9 a car purchased for \$39 446 |
| 10 a table and chairs for \$742.50 | 11 a lounge suite purchased for \$1 931.60 |
| 12 a magazine purchased for \$8.25 | 13 a TV purchased for \$638. |
- 14** Calculate the VAT on a computer valued at \$1795, if it was purchased in the following countries:
- | | |
|--------------------|--------------------|
| i) Poland | ii) Denmark |
| iii) France | iv) Spain |

In many circumstances an overseas traveller can receive a refund for the GST or VAT paid on goods. Calculate the refund possible in these scenarios.

- 15** A traveller leaving Australia with a camera purchased for A\$1356.96
- 16** A traveller leaving Australia with a video recorder purchased for A\$986.15
- 17** A traveller leaving Australia with a laptop computer purchased for A\$3007.95
- 18** A traveller leaving Norway with a statue purchased for A\$1158.16
- 19** A traveller leaving Estonia with a rug purchased for A\$770.54

Flat rate loans use the original principal, or the amount borrowed, to calculate the interest. The simple interest formula is used to calculate the interest.

e.g.

- 1** Zara borrows \$7500 over 7 years at a flat interest rate of 7.5% p.a. Calculate

- i) the total interest

$$I = 7500 \times 0.075 \\ = \$3937.50$$

The simple interest formula is $I = Prn$ where:

I = interest

P = principal (amount borrowed)

r = interest rate (usually as a decimal).

- ii) the total amount to be repaid over the 7 years

$$\text{Total to repay } I = \$7500 + \$3\,937.50 \\ = \$11\,437.50$$

- iii) the equal monthly repayments.

$$\text{Repayment} = \$11\,437.50 \div (7 \times 12) \\ = \$136.16$$

- 2** Cameron borrowed \$8500 for a motorbike. He takes out a loan, which charges a flat interest rate. If he takes the loan out over 5 years and is charged \$3612.50 in interest over this period, find the interest rate he was charged.

Using $I = Prn$

$$3612.5 = 8500 \times r \times 5 \quad \text{where } r \text{ is the decimal rate}$$

$$r = \frac{3612.5}{8500 \times 5} \\ = 0.085$$

The interest rate charged was 8.5% p.a.

For questions 1–10, calculate:

- i) the interest charged
- ii) the total amount to be repaid
- iii) the amount of equal monthly instalments.

- 1 Dave borrowed \$12 500 for a car at a flat rate of interest of 7.5% p.a. over 5 years.
- 2 Sue borrowed \$4200 for a computer at a flat rate of 10% p.a. over 3 years
- 3 Jasmine borrowed \$5200 for a sound system at a flat rate of 12% p.a. over 6 years.
- 4 Carson borrowed \$15 750 for a boat at a flat rate of 10% p.a. over 6 years.
- 5 Jade borrowed \$4700 for a holiday at a flat rate of 10.5% p.a. over 4 years.
- 6 Billie borrowed \$7420 to buy a new lounge suite at a flat rate of 9.75% over 10 years.
- 7 Benji borrowed \$10 400 for musical instruments at a flat rate of 14.3% p.a. over 8 years.
- 8 Sonny borrowed \$4876 for sporting equipment at a flat rate of 11.4% p.a. over 4 years.
- 9 Hazzan borrowed \$12 470 for a new motorbike at a flat rate of 13.3% p.a. over 7 years.
- 10 Brittany borrowed \$6070 for an overseas holiday at a 10.6% flat interest rate over 6 years.
- 11 Sarah borrows \$10 750 to buy furniture. She takes out a loan at a flat interest rate of 8.5% p.a. She pays the loan in equal monthly instalments over 5 years. Determine the monthly instalments.
- 12 Amelia borrows \$7850 for a holiday. She agrees to pay it over 5 years at a flat rate of 10.2% p.a. Find her monthly instalments.

- 13 Hilton wants to borrow \$12 000. He shops around and obtains 4 options. Which option should Hilton choose so that his interest payment is the lowest? What would be the monthly repayment on that loan?
- i) a simple interest rate of 6.5% p.a. over 8 years
 - ii) a simple interest rate of 7.4% p.a. over 7 years
 - iii) a simple interest rate of 8.6% p.a. over 6 years 3 months
 - iv) a simple interest rate of 11.3% p.a. over 4½ years
- 14 Bridget takes out a loan for some home improvements. She borrows \$12 500 and repays it off over 5 years. If she is charged \$4687.50 in interest, determine the interest rate she is charged.
- 15 Alan borrows \$6500 at a flat interest rate. Over 4 years, the term of the loan, he is charged a total of \$3900 in interest. Determine the interest rate he is charged per annum.
- 16 Freda is charged \$2980 in interest on a loan of \$7450. How many years does it take her to repay the loan if she is charged a flat interest rate of 8% p.a?
- 17 On a loan of \$7500, Ned repays a total of \$10 125 over 5 years. Determine the flat interest rate he was charged.
- 18 Alexis borrowed \$9450. She repaid the loan in 4 years. The total she repaid is \$13 230. Determine the flat interest rate she was charged.
- 19 Hugo borrows \$12 740. He is charged a flat interest rate of 8.5% p.a. and pays \$3790.15 in interest. How many years did it take to repay the loan?
- 20 Emily borrows \$2500 and repays it in 18 months at a flat interest rate of 8.25% p.a. Calculate the interest she pays.
- 21 Louise borrows \$4500. She repays the loan after 30 months. She is charged a flat rate of 9.75% p.a. Calculate the interest she pays.
- 22 Jarred repays a loan of \$4840 over 21 months. He is charged a flat rate on 10.15% p.a. How much does he pay altogether to repay the loan?
- 23 Pixie repays a loan over 5 years. She borrowed \$5250 and repays the loan at \$115.94 per month. Determine the annual flat rate of interest she is charged (to one decimal place)
- 24 Andrew borrows \$17 560 at a flat interest rate of 9.5% p.a. and agrees to pay off the loan in 5 years. Prior to the first instalment, he decides to pay the car off over 4 years instead. How much extra per month must he pay to do this?
- 25 Dan borrows \$5450 at a flat interest rate of 12% p.a.
- i) How many years would it take him to repay the loan if the interest charged is the same as the principal, and he pays the loan off at \$109 per month?
 - ii) How long would it take him to repay the loan if the repayments stay the same but the interest charged is 80% of the principal?
- 26 Paris needs \$21 350 to buy a new car. She enquires at a financial institution and they offer her 10.6% flat interest rate over 5 years.
- i) What would be her monthly repayments?
 - ii) If her parents offer her \$3500 towards the cost of the car to reduce the amount she needs to borrow, what difference to her monthly repayments would this make?

An **annuity** is an investment where equal amounts are invested in an account at the end of a given period (e.g. a month), interest compounding at the end of each of these time periods.

The **future value** of an annuity is the total value of the investment at the end of the last period. The future value (A) may be calculated using the formula:

$$A = M \left\{ \frac{(1+r)^n - 1}{r} \right\}$$
 where, M = the contribution per period, paid at the end of the period
 r = the interest rate per time period, expressed as a decimal
 n = the number of time periods

e.g. **3 1** Frederick invests \$1 500 in an annuity at the end of each year. If he invests it at 7.5% p.a., determine the value of the annuity after 6 years.

$$A = 1500 \left\{ \frac{(1+.075)^6 - 1}{0.075} \right\}$$

$$= \$10\,866.03$$

2 Mary wants to have \$6000 in 4 year’s time. She invests an amount at the end of each year in an annuity at a rate of 8% p.a. How much must she invest each year if she is to reach her goal?

$$6000 = M \left\{ \frac{(1+.08)^4 - 1}{0.08} \right\}$$

$$M = 6000 \div \left\{ \frac{(1+.08)^4 - 1}{0.08} \right\}$$

$$= \$1331.52$$

1 Dave invests in an annuity at the end of each year for 5 years. Determine the value of the annuity if the interest rate is 9% p.a. To understand the process involved in calculating annuities, complete the table below, using the compound interest formula:

Number of payments	Amount invested	Years of investment	Amount at end of term	Amount
1	\$3000	4	$3000(1.09)^4$	i).....
2	\$3000	3	$3000(1.09)^3$	ii).....
3	\$3000	2	iii).....	iv).....
4	\$3000	1	v).....	vi).....
5	\$3000	0	vii).....	viii).....
			TOTAL	ix).....

- 2 Use a table, as in question 1, to determine the future value of an annuity if \$7000 is invested at the end of each year at 6% p.a. for 4 years.

The table following gives details of five annuity investments. Calculate the future value of each annuity. In all cases the amount is contributed at the end of each year.

	Amount (\$)	Number of years	Interest rate (% p.a.)
3	2000	5	6
4	3800	6	7.5
5	7000	6	8.3
6	600	8	9.8
7	2000	20	11.1

- 8 Raphael invests \$200 in an annuity at the end of each month. If he invests it for 7 years at an annual interest rate of 15%, find the final value of the annuity. (Hint: $r = 15 \div 12$; $n = 7 \times 12$)
- 9 If you make a contribution of \$300 at the end of each month in an annuity, at an interest rate of 9% p.a., determine the final pay out you would receive at the end of 6 years.
- 10 Determine the future value of an annuity if \$350 is invested at the end of each month for 8 years, given the interest rate is 12% p.a.
- 11 Nerida invests \$500 at the end of every 6 months. If the interest rate is 10% p.a., calculate the final value after 8 years.
- 12 Nassar wishes to retire in 14 years time. He needs to have saved \$200 000. Nassar contributes to an annuity at the end of each year at an interest rate of 8% p.a. How much per year would he have to invest to enable him to reach his goal?
- 13 Suzie wants to buy a home in 5 years time. She needs a deposit of \$50 000. She decides to invest \$7500 in an annuity at the end of each year, paying 11% p.a. Will she have sufficient at the end of the 5 years? If not, what is the shortfall?
- 14 I need \$6500 in 7 years time. If I invest an amount into an annuity at the end of each month, paying 12% p.a., what monthly contribution must I make to reach this amount?
- 15 Abby invests \$1200 in an annuity at the end of each year for 6 years at 6% p.a.
Bobby invests \$100 in an annuity at the end of each month for 6 years at 6% p.a.
Which investment gives the best return, and by how much?
- 16 David has the choice of investing an amount at 12% p.a. He can invest
i) \$2400 at the end of each year for 12 years, or
ii) \$200 at the end of each month for 12 years
Which investment is the better and by how much?
- 17 Ellie and Sarah retire on the same day. Ellie has contributed to an annuity for 15 years and Sarah has been contributing for 20 years. If they both contributed \$1500 at the end of each year, at the same interest rate of 9% p.a. how much more is Sarah's lump sum?

The **present value** of an annuity is the sum of money that needs to be invested today, at the same interest rate as an annuity, so that at the end of the investing period the same amount of money is obtained.

e.g. An annuity where \$4000 is invested at 10% p.a. at the end of each year for 5 years, gives

$$\$24\,420.40 \quad \left(\text{Using } A = M \left\{ \frac{(1+r)^n - 1}{r} \right\} \right)$$

When \$15 163.15 is invested for 5 years at a compound interest rate of 10% p.a. it gives \$24 420.40 (using $A = P(1+r)^n$).

Thus \$15 163.15 is the present value of the annuity

To determine the present value of an annuity we use:

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\} \quad \text{or} \quad N = \frac{A}{(1+r)^n}$$

Where A = present value of the annuity, M = contribution at the end of each period
 n = number of periods r = interest rate per period

In the example above, the Present value is calculated by: $N = 4000 \left\{ \frac{(1+.1)^5 - 1}{0.1(1+.1)^5} \right\}$
 $= \$15\,163.15$

Determine the present value of these annuities, given that in each case the contribution is invested at the end of each period.

	Contribution	Interest p.a.	Period	Number of years
1	\$3000	7%	yearly	6
2	\$4000	8%	yearly	10
3	\$2500	9%	yearly	25
4	\$3000	8.5%	yearly	40
5	\$400	12%	monthly	10
6	\$300	6%	monthly	25
7	\$1000	8%	quarterly	15
8	\$1200	12%	quarterly	12

- How much must be invested now to give the same amount as an annuity where \$1000 is invested at the end of each year for 10 years at a compound interest rate of 10% p.a.?
- Dianne invests \$3000 in an annuity at the end of each year for 10 years at an interest rate of 7% p.a. How much must Dave invest now at 10% p.a. so that he will have the same amount of money as Dianne after 10 years?
- Denis invests in an annuity paying 9.5% p.a. His contributions are \$800 at the end of each year for 12 years. Ali invests a lump sum. How much must Ali invest if he is to have the same amount as Denis after 12 years if his interest rate is also 9.5% p.a.?

- e.g. 1** Calculate the monthly repayments of a loan of \$20 000 if the interest rate is 9.6% p.a., compounding monthly, and the loan is to be repaid over 10 years. Hence calculate the total amount repaid and the amount of interest paid. (Note: use the Present Value formula.)

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\} \quad \text{where } N = 20\,000, n = 12 \times 10 = 120, r = 0.096 \div 12 = 0.008$$

and M = monthly repayments

$$20\,000 = M \left\{ \frac{(1.008^{120} - 1)}{0.008(1.008^{120})} \right\}$$

$$M = 20\,000 \div \left\{ \frac{1.008^{120} - 1}{0.008(1.008^{120})} \right\}$$

$$= \$259.89$$

The monthly repayment is \$259.89.

$$\begin{aligned} \text{Amount repaid} &= \$259.89 \times 120 \\ &= \$31\,186.80 \end{aligned}$$

$$\begin{aligned} \text{Total interest paid} &= \$31\,186.80 - \$20\,000 \\ &= \$11\,186.80 \end{aligned}$$

- 2** Don wants to take 12 months off work to travel. He needs to have a regular monthly income of \$2100. Calculate the lump sum he needs to invest in order to do this, if the interest rate is 10.8% p.a. compounding monthly.

Use the present value formula, where $r = 0.108 \div 12 = 0.009$ $M = 2100$ $n = 12$

$$N = 2100 \left\{ \frac{(1.009)^{12} - 1}{0.009(1.009)^{12}} \right\}$$

$$= \$23\,785.69$$

Don needs to invest \$23 785.69.

- 1** Complete the working below to determine the monthly repayment on a loan of \$300 000 if the interest rate is 10.8% p.a. compounded monthly and the loan is to be repaid over 5 years.

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\} \quad \text{where i) } N = \dots\dots\dots \quad \text{ii) } n = \dots\dots\dots \quad \text{iii) } r = \dots\dots\dots \div 12 = \dots\dots\dots$$

and M = monthly repayments

$$\text{iv) } \dots\dots\dots = M \left\{ \frac{(1.009)^{\text{(v)} \dots\dots\dots} - 1}{\text{(vi) } \dots\dots\dots (\text{(vii) } \dots\dots\dots)^{60}} \right\}$$

$$M = \text{(viii) } \dots\dots\dots \div \text{(ix) } \dots\dots\dots$$

$$M = \text{(x) } \dots\dots\dots$$

xi) Hence calculate the total amount repaid and the amount of interest paid.

- 2** Delta wants to have time off work to care for her children. Complete the working below to determine (to the nearest \$) how much she must invest at 12% p.a. so that she will have an amount of \$1000 a month for 4 years.

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\} \quad \text{where i) } n = \dots\dots\dots \quad \text{ii) } r = \dots\dots\dots \div 12 = \dots\dots\dots \quad \text{iii) } M = \dots\dots\dots$$

and N = amount to be invested

$$N = \text{iv} \dots\dots\dots \left\{ \frac{((\text{vi})\dots\dots\dots)^{48} - 1}{(\text{vi})\dots\dots\dots((\text{vii})\dots\dots\dots)^{48}} \right\}$$

$$N = (\text{viii})\dots\dots\dots$$

Determine the monthly repayment on each of these housing loans, given that interest is calculated monthly.

- 3 A loan of \$120 000 at 12% p.a. to be repaid over 25 years.
- 4 A loan of \$250 000 at 9.6% p.a. to be repaid over 25 years.
- 5 A loan of \$18 000 at 8.4% p.a. over 20 years.
- 6 A loan of \$175 000 at 7.2% p.a. over 30 years.
- 7 A loan of \$340 000 at 4.8% p.a. over 35 years.
- 8 Determine the monthly repayments on the following loans, where interest is calculated monthly. Hence calculate the total amount to be repaid and the total interest paid.
 - i) A loan of \$200 000 at 9% p.a. over 20 years
 - ii) A loan of \$21 000 at 9.3% p.a. over 15 years
- 9 Jarrod and Jade both take out housing loans of \$200 000 each, at an interest rate of 6% p.a., interest compounding monthly. Jarrod repays his loan over 22 years, and Jade over 18 years.
 - i) Calculate Jarrod's and Jade's monthly payments.
 - ii) How much does Jade save by paying her loan off in 18 years, when compared to Jarrod?
- 10 Zara and Carson each borrow \$250 000 for a home. They both decide to make monthly repayments over 28 years. Zara pays an interest rate of 7.2% p.a. monthly reducible, and Carson 7.5% p.a. monthly reducible. How much less will Zara pay over the period of the loan?
- 11 Ruth's parents offer to invest some money so that for the next 10 years her income will be increased by \$900 a month. How much money must be invested at a rate of 9.9% p.a. compounded monthly?
- 12 Maisie decides to take 4 years off work to care for her children. She will require an income of \$1000 a month for this period. How much will she need to invest now, at 8.4% p.a. compounded monthly, if she is to achieve this?
- 13 Alicia's parents intend to invest an amount of money to ensure that in the 5 years she is at university she will receive an income of \$600 a month. How much must her parents invest at 6% p.a. compounding monthly?
- 14 Austin decides to take 3 years off work to complete his studies. He decides to invest a lump sum at 6.6% p.a. How much should he invest to give him a monthly income of \$800 if the interest rate is calculated monthly?
- 15 Albert wishes to retire. He wants to take out an annuity that pays \$2800 a month for 20 years.
 - i) Determine the lump sum he needs to invest if the annuity pays an interest rate of 9% p.a. compounding monthly.
 - ii) How much more would Albert need to invest if he wanted to receive an annuity which pays him \$3000 a month?
- 16 John wants to retire. He decides he needs \$10 000 every 6 months for 25 years. How much should John invest to reach his goal if interest is 9.4% compounded half yearly?
- 17 Rita wants to retire. She takes out an annuity that pays \$2000 a month for 25 years. How much must she invest if she receives an interest rate of 8.4% p.a. compounded monthly?

You will need the following formulae:

$$I = Prn$$

$$A = P(1 + r)^n$$

$$A = M \left\{ \frac{(1 + r)^n - 1}{r} \right\}$$

$$N = M \left\{ \frac{(1 + r)^n - 1}{r(1 + r)^n} \right\}$$

where:

I = simple interest

P = Principal

r = rate of interest per period

n = number of periods

A = future value

M = contribution per period

N = present value

- 1 The GST in Australia is 10%. How much GST must be added to each of the following before sale?
 - i) a \$3485 television
 - ii) a \$15 350 car
 - iii) a \$310 bike
 - iv) a \$95 ironing board
- 2 GST in New Zealand is 12.5%. Determine the New Zealand GST payable on the four items in question 1.
- 3 Find the retail price of a car bought in Australia, valued at \$45 380 before GST.
- 4 In Australia, GST on an item can be found by dividing the total price by 11. Determine the GST paid on:
 - i) a sound system priced at \$2 043.80
 - ii) a gas bill of \$201.63
- 5 Use the unitary method to determine the GST paid if you purchased the following items in New Zealand:
 - i) a television with a purchase price of \$3898.17
 - ii) a refrigerator for \$1856.25.
- 6 André borrows \$10 500 at a flat interest rate of 6.8% p.a. and repays the loan over 6 years. How much interest will he have to pay?
- 7 Alicia borrows \$15 800 to buy a car, at a flat rate of interest of 11.5% p.a. Determine the equal monthly repayments if she repays the loan in 5 years.
- 8 Sophia borrows \$5400 and repays it over 6 years. Determine the simple interest rate charged if the monthly repayments were \$111.
- 9 Kamal invests \$1500 at the end of each year in an annuity that pays an interest rate of 7% p.a. How much will Kamal have at the end of 15 years?
- 10 Rachael invests in an annuity that pays an interest rate of 7.2% p.a. How much will Rachael have after 20 years, if she invests \$17 300 at the end of each year?
- 11 Fred invests \$150 at the end of each month for 21 years. How much will he receive if the interest rate is compounded monthly at the rate of 8.4% p.a.?
- 12 Jasmine needs \$32 000 in 12 years time. How much must she invest at the end of each year at an interest rate of 10% p.a. compounding annually to reach her goal?
- 13 Determine the present value of an annuity when \$1500 is invested at the end of each year for 7 years at 9% p.a.
- 14 Michael estimates that he will need \$1 100 000 when he retires in 30 years. He decides to invest \$8000 at the end of each year, compounding annually at a rate of 9.5% p.a. Will he reach his goal? If so, by how much will he be over? If not how much will he be short of his target?

e.g.

- 1 BHP Billiton declared a dividend of 36.35 cents per share. Determine the net dividend yield as a percentage of a share price of \$20.83.

$$\begin{aligned}\text{Share net \% yield} &= \frac{0.3635}{20.83} \times 100\% \\ &= 1.745\%\end{aligned}$$

- 2 The price of CSR shares is \$2.67. The net dividend yield is 4.49%. Find the dividend per share.

$$\begin{aligned}\text{Dividend} &= \$ (0.0449 \times 2.67) \\ &= \$0.11988\end{aligned}$$

The dividend per share is 12 cents.

Shares may be purchased through a traditional sharebroker or a discount brokerage firm
Note: GST of 10% is only charged on the brokerage and not on the cost of the shares.

The tables below give a typical charge for both types of brokerage. Students should familiarise themselves with the advantages and disadvantages of the different types of brokerage.

Traditional brokerage

Up to \$5000	\$75
Up to \$10 000	1.5%
Up to \$30 000	1.25%
Up to \$50 000	1%
Above \$50 000	0.8%

Discount brokerage

On-Line		Over the Phone	
Up to \$10 000	\$29.95	Up to \$10 000	\$ 54.60
Above \$10 000	0.13%	Up to \$15 000	0.54%
		Up to \$ 80 000	0.45%
		Up to \$1m	0.40%

- 1 The Seven network pays a dividend of 23.5 cents per share. Determine the net percentage yield on a share price of \$8.15.

Given the following company data, determine, for each company dividend, the net yield as a percentage (to two decimal places) of the share price:

- 2 Mayne Group – a dividend of 11.1c per share and a share price of \$5.04
- 3 NAB – a dividend of 166c per share and a share price of \$30.99
- 4 Origin Energy – a dividend of 15c per share and a share price of \$7.14
- 5 Santos – a dividend of 36c per share and a share price of \$11.52.
- 6 Stockland – a dividend of 38.9c per share and a share price of \$6.00.
- 7 A.M.P. – a dividend of 37c per share and a share price of \$9.33.
- 8 A.S.X. – a dividend of 120c per share and a share price of \$34.85.

Determine the dividends (in cents to two decimal places) for each company, given that:

- 9** The dividend yield for Westpac shares is 4.66% on a share price of \$19.95.
- 10** The dividend yield for Woolworth shares is 3.16% on a share price of \$16.14.
- 11** The dividend yield for Booral shares is 4.32% on a share price of \$7.87.
- 12** The dividend yield for John Fairfax shares is 4.00% on a share price of \$4.62.
- 13** The dividend yield for the Rinker Group shares is 1.38% on a share price of \$15.26.
- 14** The dividend yield for the Coles shares is 3.1% on a share price of \$13.30.
- 15** The dividend yield for the ANZ Bank shares is 4.11% on a share price of \$27.80.
- 16** David buys 900 St George Bank shares for \$21.36. During the year the bank pays a dividend of 17.9 cents per share. At the end of the year the shares are selling for \$27.15.
 - i)** Determine the cost of buying the shares (do not include brokerage).
 - ii)** Determine the dividend received.
 - iii)** Determine the value of the shares at the end of the year including dividends.
- 17** Farah buys 1500 Ten Network shares for \$4.25 a share. The dividend paid per share is 24c. At the end of the year, the shares are selling for \$3.70. Calculate:
 - i)** the initial cost of the shares (do not include brokerage)
 - ii)** the total dividend paid
 - iii)** the value of the shares (including the dividends at the end of the year).
- 18** David and Katherine each have \$10 032 to buy shares (excluding GST and brokerage). David buys Macquarie Bank shares for \$66.00 a share. Katherine buys Macquarie Airport shares for \$3.30 a share. If Macquarie Bank pays a dividend of 161 cents a share and Macquarie Airport pays a dividend of 19 cents per share, determine who earns the greater return on their money and by how much.

For each of the following, determine:

- i)** the total cost of buying the shares, including brokerage and GST
- ii)** the amount received when the shares are sold, after all costs have been deducted.
- iii)** the profit/lost made.
- 19** 2500 CBA shares bought for \$30.05 a share, and sold 5 years later for \$34.19 using discount brokerage (on line).
- 20** 3000 NAB shares bought at \$31.95 a share and sold 2 years later for \$30.02, using traditional brokerage.
- 21** 2500 shares bought in Telstra for \$7.50 a share. Sold 3 years later, using a discount broker (over the phone), for \$5.50 a share
- 22** Pat wants to receive at least \$500 a year in dividends.
 - i)** If she decides to buy Booral shares, which pay a dividend of 42c per share, what is the minimum number of shares she needs to purchase?
 - ii)** How much will it cost her, including GST and discount brokerage (on-line), if the shares are \$7.73 each?

One method of determining the number of periods required to repay a loan, given the amount borrowed, the periodic repayments and the interest rate, is to use the "guess and check" method

e.g. Determine the number of repayments required to repay a loan of \$250 000 at a monthly reducible rate of 6% p.a. if a monthly repayment of \$1499 is made.

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\} \quad \text{where: } N = \text{amount borrowed}$$

$$M = \text{monthly repayments}$$

$$n = \text{number of repayment periods}$$

$$r = \text{interest rate per period (as a decimal)}$$

$$250\,000 = 1499 \left\{ \frac{1.005^n - 1}{0.005(1.005)^n} \right\}$$

Guess $n=28$ years (i.e. $28 \times 12 = 336$ repayments)

$$\begin{aligned} \text{R.H.S.} &= 1499 \left\{ \frac{1.005^{336} - 1}{0.005(1.005)^{336}} \right\} \\ &= \$243\,690.69 \quad \text{which is less than } \$250\,000 \end{aligned}$$

Guess $n=29$ years (348 repayments)

$$\begin{aligned} \text{R.H.S.} &= 1499 \left\{ \frac{1.005^{348} - 1}{0.005(1.005)^{348}} \right\} \\ &= \$246\,950.35 \quad \text{which is also less than } \$250\,000 \end{aligned}$$

Guess $n=30$ years gives \$250 020.36 which is very close to \$250 000

The number of periods is 360 months or 30 years.

Use the guess and check method to determine the approximate number of repayment periods on each loan. The given interest rates are rates p.a. and they are monthly reducible.

- 1 \$100 000 invested at 7.2% with monthly repayment of \$787.
- 2 \$170 000 invested at 6% with monthly repayments of \$1095.
- 3 \$250 000 invested at 8.4% with monthly repayments of \$2447.
- 4 \$140 000 invested at 12% with monthly repayments of \$2008.59.
- 5 \$62 000 invested at 9.6% with monthly repayments of \$629.
- 6 \$145 050 invested at 9% with monthly repayments of \$1224.
- 7 \$98 000 invested at 15% with monthly repayments of \$1570.
- 8 \$75 000 at 6% with monthly repayments of \$731.80.
- 9 \$105 000 at 9.6% with monthly repayments of \$1325.20.
- 10 \$68 000 at 7.2% with monthly repayments of \$719.30.
- 11 \$120 000 at 6% with monthly repayments of \$859.72.
- 12 \$250 000 at 6.6% with monthly repayments of \$1703.67.
- 13 \$310 000 at 7.2% with monthly repayments of \$2104.24.
- 14 \$89 000 at 9.3% with monthly repayments of \$848.67.

When the value of an asset decreases it is said to have depreciated. Examples of assets, which depreciate, are cars, computers, machinery, televisions, etc.

There are two methods of calculating depreciation:

1 Straight line method

$$S = V_0 - Dn$$

where S = salvage (current) value of the asset, D = amount of depreciation per period
 V_0 = purchase price of the asset, n = total number of periods
 r = percentage depreciation per period

2 Declining Balance Method

$$S = V_0(1-r)^n$$

e.g. A company purchases a welding machine for \$10 000. Determine the value of the machine after 3 years by:

- | | |
|--|------------------------|
| i) the straight line method, given that it depreciates \$1500 a year | |
| ii) the declining balance method given that the rate of depreciation is 15%. | |
| i) $S = V_0 - Dn$ | i) $S = V_0(1-r)^n$ |
| $S = \$ (10000 - 1500 \times 3)$ | $= \$ [10000(0.85)^3]$ |
| $= \$5500$ | $= \$6141.25$ |

Determine the salvage value of each item using the straight-line method.

- 1 A television bought for \$3500, depreciating at \$400 per year for 5 years
- 2 A tractor bought for \$54 000, depreciating at \$7030 per year for 7 years
- 3 A computer valued at \$2350, depreciating \$510 per year for 4 years
- 4 A drilling machine, valued at \$8550, depreciating at \$1400 per year for 5 years
- 5 A caravan valued at \$23 500, depreciating at \$2200 per year for 7 years.

Determine the salvage value of the each item using the declining balance method.

- 6 A car bought for \$45 000 depreciating at a rate of 21% p.a. for 4 years.
- 7 A welding machine bought for \$34 000 depreciating at a rate of 33% p.a. for 5 years.
- 8 A television valued at \$9450 depreciating at a rate of 25% p.a. for 3 years.
- 9 A laptop computer bought for \$3400 depreciating at a rate of 29% p.a. for 4 years.
- 10 A truck purchased for \$150 000, depreciating at a rate of 28% p.a. for 7 years.
- 11 Fred purchased a \$70 000 machine and depreciated it using the declining balance method at a rate of 30% p.a.
 Dave purchased the same machine for the same price and depreciated it using the straight-line method at a rate of \$10 000 a year.
 Complete the table below and draw a graph to represent both methods. Determine when the depreciated values are the same.

Years	0	1	2	3	4	5	6	7
Fred	\$70 000	\$49 000	\$.....	\$24 010	\$16 807	\$.....	\$8235	\$.....
Dave	\$70 000	\$60 000	\$.....	\$40 000	\$30 000	\$.....	\$10 000	\$.....

- 12 A car is valued at \$34 000. After one year using the declining balance method, it is valued at \$23 800. Determine the percentage rate of depreciation, and its value after 3 years.
- 13 A truck is purchased for \$75 000. After 2 years it has depreciated to \$51 000 using the straight-line method. Determine the book value after 6 years.
- 14 A car is bought for \$45 000. After 1 year its value is \$27 000, using the declining balance method. What is its value after 4 years?
- 15 A piece of machinery was purchased for \$150 000. After 3 years, using the straight-line method, its value was \$97 500. What was its value after 5 years?

The following questions refer to the **bill no. 5 – aussie visa credit card**.

- 1 What was the balance from the previous statement?
- 2 Was it paid in full?
- 3 What do the terms AU and CR mean?
- 4 How much must be paid to completely pay this account?
- 5 When was the payment due?
- 6 What is the minimum amount that can be paid?
- 7 Show that the minimum payment is 2.5% of the closing balance.
- 8 If the closing balance was \$1434.80, calculate the minimum payment.
- 9 What is the limit on this credit card?
- 10 Show that the daily interest rate is the annual rate divided by 365, correct to 5 decimal places.

The following questions refer to the **bill no. 6 – aussie visa credit card**.

- 11 What was the previous balance?
- 12 How much was spent on this account?
- 13 How much must be paid in order to avoid interest charges?
- 14 How is the available credit calculated?
- 15 If you have an unauthorised transaction, what phone number do you ring?
- 16 What is the daily interest rate?
- 17 If there was a balance of \$1500 outstanding on the 1st April, and compound interest was charged until the account was paid in full, how much interest would be charged if the account was paid on 24th April? (Interest is not charged on the day of payment.)

The following questions refer to the **bill no. 7 – aussie visa credit card**.

- 18 What does "DR" mean?
- 19 Determine the closing balance.
- 20 The monthly repayment due is 2.5% of the closing balance. Calculate the monthly repayment.
- 21 Determine the available credit.
- 22 The amount spent at Woolworths has been deleted. Determine the amount.
- 23 How many days did it take for most transactions to be completed?
- 24 Was the previous outstanding balance paid in full?
- 25 How many days are covered by this statement?
- 26 Calculate the interest charged if Mr Rose owed \$600 for 29 days.
- 27 If the "annual % rate" was increased to 17.009%, determine the new "daily % rate".

Interest on outstanding balances is calculated using the compound interest formula, and is calculated daily.

- 28 Calculate the interest charged if \$1550 is outstanding for 20 days at a daily interest rate of 0.05%.
- 29 Calculate the interest charge if \$4500 is outstanding for 27 days at a daily interest rate of 0.05%.
- 30 Calculate the interest charged, at a daily rate of 0.04%, if \$3150 is outstanding on the 1st February and the bill is paid in full on 28th February.
- 31 By first calculating the daily interest rate, calculate the interest charged on \$3450 outstanding for 25 days at an annual interest rate of 14.6%.

You will need the following formulae:

$$A = P(1+r)^n$$

$$S = V_0(1-r)^n$$

$$A = M \left\{ \frac{(1+r)^n - 1}{r} \right\}$$

$$N = M \left\{ \frac{(1+r)^n - 1}{r(1+r)^n} \right\}$$

where:

P = Principal

r = rate of interest per period

n = number of periods

A = future value

M = contribution per period

N = present value

S = salvage value after a period

- 1 Monica travels overseas. If she claims the GST back on a camera that cost \$792 (including 10% GST), determine her refund.
- 2 Raymond is charged a daily interest rate of 0.0456% (compounded daily) on the outstanding balance of his credit card account. If the balance was \$850 and it was outstanding for 28 days, calculate (to the nearest cent) the interest that he is charged.
- 3 Given that 1 Australian dollar is worth 28 Thai Baht and 1 Indian Rupee is worth 0.8 Thai Baht, how many Indian Rupees would you get for \$250 Australian?
- 4 Bill uses the declining balance method of depreciation to calculate the depreciation on his computer. He buys a computer for \$7500 and depreciates it at a rate of 35% p.a.
 - i) How much does the computer depreciate in the first year?
 - ii) What is the value of the computer after 4 years?
 - iii) The Taxation Department allows the computer to be written off when its value is less than \$500. In what year would this occur?
- 5 The dividend yield on 'Oil Search' shares is 2.64%. If the value of each share is \$3.57, determine the dividend per share in cents (to 2 decimal places).
- 6 St George Bank pays a dividend of 144 cents per share. If the dividend yield is 4.47%, determine the value of each St George share (to the nearest cent).
- 7 Dave buys 1200 Coles Myer shares at \$10.33 per share. His brokerage charge is 1.35% of the value of the shares plus 10% GST on the brokerage.
 - i) How much does it cost Dave to buy his shares?
 - ii) Two weeks later Dave sells them for \$13.53 per share. If the broker charges the same, 1.25% plus GST, how much profit does Dave make?
- 8 Each week Alfred is paid \$711.45. This is made up of 37.5 hours of normal time and 6 hours of overtime at time-and-a-half. What is his normal hourly rate of pay?
- 9 Wallace earns \$895 per week. Determine her weekly net pay, after tax and Medicare levy have been deducted. (See Topic 12 for Tax and Medicare levy information.)
- 10 Use the Future Value formula to determine the value of an annuity where \$250 is invested at the end of each month for 16 years at a rate of 15% p.a. compounding monthly.
- 11 Determine the monthly repayments on a loan of \$150 000 if the interest rate is 10.8% p.a., calculated monthly, and the loan is repaid over 10 years. Hence calculate the total amount repaid and the amount of interest paid.
- 12 Amelia wishes to take 4 years off work to travel. She needs an income of \$1200 per month during this time. How much must she invest to achieve her goal if the interest rate is 9% p.a. compounding monthly?

table 1: youth allowance payment rates

Status	Allowance paid per fortnight
Single, no children:	
Under 18, at home	\$183.20
Under 18, away from home	\$334.70
18 and over, away from home	\$334.70
18 and over, at home	\$220.30
Single, with children	\$438.50
Partnered, no children	\$334.70
Partnered with children	\$367.50

Special rate for long-term unemployed or migrant English students 21 years or over commencing full-time study	
Status	Allowance paid per fortnight
Single, living at home	\$270.50
Single, living away from home	\$406.50
Partnered, no children	\$367.50

table 2: personal income test for youth allowance

Family situation	For full Allowance (per fortnight)***	For part Allowance (per fortnight)
Single, under 18, at home, students and New Apprentices	up to \$236	less than \$520.57
Single, under 18, at home, job seekers	up to \$62	less than \$346.57
Single, 18 and over, at home, students and New Apprentices	up to \$236	less than \$573.57
Single, 18 and over, at home, job seekers	up to \$62	less than \$399.57
Single or partnered, away from home, students and New Apprentices	up to \$236	less than \$737.00
Single or partnered, away from home, job seekers	up to \$62	less than \$563.00
Partnered, with dependants, students and New Apprentices	up to \$236	less than \$783.86
Partnered, with dependants, job seekers	up to \$62	less than \$609.86
Single, with dependants, students and New Apprentices	up to \$236	less than \$885.29
Single, with dependants, job seekers	up to \$62	less than \$711.29
Single, long-term unemployed aged 21 or over undertaking full-time study or New Apprenticeship, at home	up to \$236	less than \$645.29
Single, long-term unemployed aged 21 or over undertaking full-time study or New Apprenticeship, away from home	up to \$236	less than \$839.57
Partnered, long-term unemployed aged 21 or over undertaking full-time study or New Apprenticeship, no dependants	up to \$236	less than \$783.86

**For job seekers, fortnightly income between \$62 and \$142 reduces fortnightly allowance by 50 cents in the dollar. Income above \$142 reduces payment by 70 cents in the dollar. For students and New Apprentices, fortnightly income between \$236 and \$316 reduces fortnightly allowance by 50 cents in the dollar. Income above \$316 reduces payment by 70 cents in the dollar. For students and New Apprentices the income test is subject to income bank credit. Partner income which exceeds cut-out point reduces fortnightly allowance by 70 cents in the dollar.

table 3: age pension payment rates

Status	Maximum Pension Rate Per Fortnight
Single	\$499.70*
Couple	\$417.20* (each)

table 4: income test for aged pensioners

Family situation	For full payment (per fortnight)*	For part payment (per fortnight)
Single	up to \$124	less than \$1387.75
Single plus 1 child	up to \$148.60	less than \$1412.35
Couple (combined)	up to \$220	less than \$2320.50
Illness separated couple (combined)	up to \$220	less than \$2747.50
Additional children	add \$24.60 per child	

*Income over these amounts reduces the rate of pension payable by 40 cents in the dollar (single), 20 cents in the dollar each (for couples).

bill 1: aussie water corporation – water rates account



Mr D Rose
41 Table Rd
Port Melden 2731

Aussie Water Corporation
ABN 741 367 7
88 Prince St
Port Melden 2731

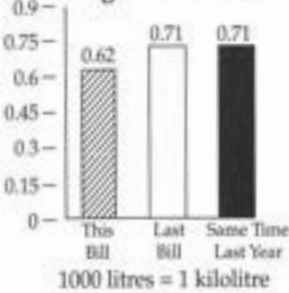
INTERNET: www.aussiewater.com.au
EMAIL US: enquiries@aussiewater.com.au
ACCOUNT ENQUIRIES: 1300 7498
PAYMENTS BY PHONE: 131 13 Billpay Code: 123
24 hours a day/7 days week Ref: 191970

CUSTOMER NUMBER:	191970
PAYMENT DUE BY:	15/04/06
AMOUNT DUE:	\$213.80

SUMMARY OF CHARGES		AMOUNT
Balance brought forward		0.00
FIXED CHARGES	01 March 2006 - 30 June 2006	
Water Service		8.45
Sewer Service		79.78
Drainage Availability Charge		14.10
Environmental Improvement		16.31
USAGE CHARGES (see over)	1 December 2006 - 5 April 2006	95.16
TOTAL AMOUNT DUE		\$213.80

Date read	Meter Reading	Water Used (kL)	Number of Days	Your Daily Average
05/04/06	866	78	125	0.62

Average daily Water Usage in Kilolitres



Your Usage Charges for the period 1 December 2005 to 5 April 2006 are made up of:
Water Usage of 78kL at 101 cents per kilolitre = \$78.78
Sewer Usage of 39kL at 42 cents per kilolitre = \$16.38

Sewer Use is determined by a Discharge Factor. 50% of the 78 kL of water used (39 kL) is deemed to have been discharged to the sewer system for transportation, treatment and disposal.

DIRECT DEBIT
Call 1300 111222 for an application.

Billers Code:	123
Ref:	191970

Contact your participating Bank, Credit Union or Building Society.

PAYMENT BY MAIL
Send this payment slip with your cheque made payable to:

Aussie Water
Loeue Bay 147
Sydney NSW 2192

PAY IN PERSON AT ANY POST OFFICE
(Credit cards not accepted)

Billers Code:	123
Ref:	191970

PAY BY CREDIT CARD
Telephone 13171 or visit www.postbillpay.com.au
Bankcard, MasterCard and Visa only.
(Credit card maximum payment is \$5,000)

bill 2: aussie city council – rates & charges notice

RATES & CHARGES NOTICE

Aussie City Council
48 Charles St
Port Melden 2731

Mr D Rose
41 Table Rd
Port Melden 2731

ACCOUNT NUMBER

67948245

For Period	01/07/06 To 30/06/07
AMOUNT DUE	\$285.09
DUE DATE 1 st Instalment	31/08/06
SERVED BY POST	20/07/06

Billers Code: 123
Ref No: 67948245

Receipt No.
Date Paid:

Please deduct any payments made after
8 July 2006

Description & Situation of Land Rated

Dp 4631 Lot 469 No 41 Table Rd Port Melden 2731

Particulars	Land Value (Base Date 1/7/01)	Cents in \$	AMOUNT
Current			
Residential Ad Valorem Rate	210 000	0.34499900	\$724.50
Residential Base Rate – 25%	1	186.60000000	\$186.60
Rivers Catchment Authority	210 000	0.01470000	\$30.87
Dom. Waste Management Serv. Charge	1	176.72000000	\$176.72
Environmental Levy-\$15 min.	210 000	0.00876400	\$18.40

The first instalment of this account will be paid via Direct Debit on 31 August 2006.

Instalment	Amount	Due Date	Total Amount Due	\$1,137.09
SECOND	\$284.00	30/11/2006	OR	
THIRD	\$284.00	28/02/2007	1 st Instalment Due	\$285.09
FINAL	\$284.00	31/05/2007	Amount Due By	31/08/2006

General Manager

NOTE: Daily interest will accrue on overdue Rates & Charges at 9.0% p.a.

Official Receipt

Bank Stamp / / \$

If mailing payment to Council and receipt required place tick in box ☐
Please read the back of this notice for information about how you can pay your rates.

bill 3: aussie energy – quarterly electricity tax invoice



www.aussieenergy.com.one

Mr D Rose
41 Table Rd
Port Melden 2731

Account Number
46 38 74 49

Direct Debited on
15 June 2006

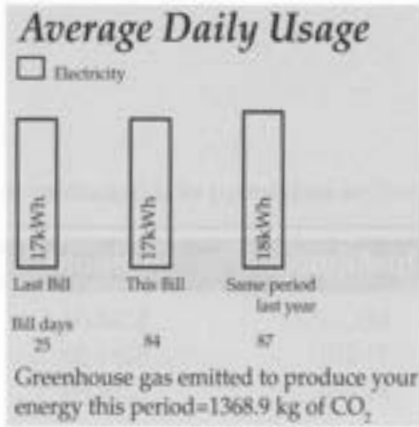
Amount Payable
\$191.27

Quarterly Electricity Tax Invoice

Location: 41 Table Rd, Port Melden 2731

Total amount payable of your last bill dated 3 March 2005	243.74
Payments – Thankyou	-243.74 CR
Electricity (04/03/2006 to 31/05/2006)	179.09
Electricity Discount: 4% (29/03/2006 to 31/05/2006)	-5.21 CR
Subtotal of charges before GST	173.88
Total GST payable 10%	17.39
Total charge including GST	191.27
Total Amount Payable	\$191.27

For Your Information



Energy Used & Costs

METER ID	THIS READING	- LAST READING	= ENERGY USED	×	USAGE SPLIT	×	RATE	=	COST
Single Energy Rate-Contract (04/03/06 - 28/03/06)									
46 38 74 49	73.0	99646.0	427.0 kWh		First	427.0	9.8880c		\$42.22
Single Energy Rate-Contract (29/03/06 - 31/05/06)									
46 38 74 49	1167.0	73.0	1,094.0 kWh		First	1,094.0	10.3000c		\$112.68
Service Availability Charge (04/03/06 - 28/03/06)			25 Days				26.3808c/Day		\$6.60
Service Availability Charge (29/03/06 - 31/05/06)			64 Days				27.4800c/Day		\$17.59
Total Electricity before GST			1521.0 kWh						\$179.09

bill 4: aussie telco – your bill

Tax Invoice – issued 25 April 06



Your Bill

Account number
789 4287
Bill number
T 684 372

Mr D Rose
41 Table Rd
Port Melden 2731

Total \$269.63
We will debit your
nominated account
on 11 Jul '06

Account Activity

Previous balance	\$253.29	
We received	\$253.29 CR	
Balance	\$0.00	
New charges pay by 11 Jul '06	\$269.63	includes rounding of \$0.01 CR
Total	\$269.63	
GST included in new charges		\$24.51

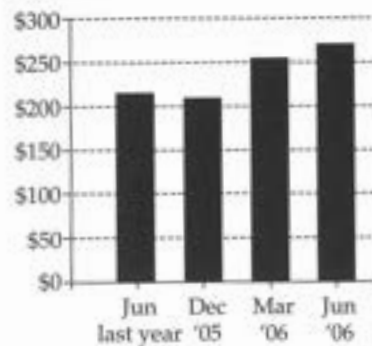
Summary of Charges

	Incl GST	
Telephone 0294731841	\$143.62	▼ \$12.11
Telephone 0294614173	\$55.76	▼ \$1.12
MobileNet 0441731695	\$24.78	▼ \$0.76
Internet and Data 46972846	\$45.48	▲ \$30.32
Total	\$269.64	▲ \$16.33

Your Benefits

Rewards Package 5%
Combining your eligible home phone with your Mobile, Pay TV or Internet services on this bill, has entitled you to a 5% discount on eligible calls and subscription charges for those two services.

Compare your bill totals



bill 5: aussie visa – credit card monthly statement



Mr D Rose
41 Table Rd
Port Melden 2731

Statement for
Aussie Visa
ABN 473 685 291

Your Account Summary

Balance from Previous Statement	\$653.33 DR
Payments and other credits	\$661.27 CR
Purchases, cash advances	\$652.77 DR
Interest and other charge	\$0.00
Closing Balance	\$644.83 DR
Monthly Payment Due By 23 December 2006	\$16.12
Minimum Payment	\$16.12

Statement Period: 11 November 2006 to 10 December 2006
Account No: 6554 4122 9734
Credit Limit: \$6,000
Available Credit: \$5,355
Page No: 1

Closing Balance Breakdown

Transaction Type	Annual % Rate	Daily % Rate	Balance
Purchase	16.650	0.04561	\$644.83

Your Transaction Record

Date Processed	Date of Transaction	Details	Amount As
14/11/06	13/11/06	OPTUS MOBILE AUTOPAY NORTH SYDNEY	\$77.00
14/11/06	14/11/06	ONLINE DIRECT ROWVILLE AU	\$6.27
15/11/06	13/11/06	MILBANK JEWELLERY MELBOURNE	\$55.00
17/11/06	17/11/06	DAVID JONES BONDI JUNCTION AU	\$79.88
20/11/06	17/11/06	ENERGY AUST AUTOPAY SYDNEY	\$197.07 CR
20/11/06	20/11/06	CASH PAYMENT THANK YOU MELBOURNE	\$50.00 CR
21/11/06	20/11/06	WORLD VISION BURWOOD EAST	\$39.00
21/11/06	20/11/06	SEAVIEW GALLERY MIRANDA AU	\$85.00
22/11/06	22/11/06	OPTUS WIRELINE A/PAY NORTH SYDNEY	\$2.37
22/11/06	20/11/06	MYER SYDNEY	\$20.00 CR
23/11/06	23/11/06	NRMA SYDNEY AU	\$33.48
24/11/06	24/11/06	CASH PAYMENT THANK YOU MELBOURNE	\$605.00
04/12/06	03/12/06	GIO/SML SYDNEY	63.97

Unauthorized or Unknown Transactions: To report an unauthorized or unknown transaction, please call 13 476. It is in your interest to report any unauthorized or unknown transactions immediately, as a delay in notification may limit the Aussie ability to investigate the transactions.

For your
record

Date Paid	Chq/Rec no.	Amount \$
-----------	-------------	-----------

bill 6: aussie visa – credit card monthly statement



Mr D Rose
41 Table Rd
Port Melden 2731

Statement for
Aussie Visa
ABN 473 65 291

Your Account Summary

Balance from Previous Statement	\$277.34 DR
Payments and other credits	\$542.35 CR
Purchases, cash advances	\$877.02 DR
Interest and other charge	\$0.00
Closing Balance	\$612.01 DR
Monthly Payment Due By 23 September 2006	\$15.30
Minimum Payment	\$15.30

Statement Period: 11 August 2006 to 9 September 2006

Account No: 6554 4122 9734

Credit Limit: \$6,000

Available Credit: \$5,387

Page No: 1

Closing Balance Breakdown

Transaction Type	Annual % Rate	Daily % Rate	Balance
Purchase	16.650	0.04561	\$612.01

Your Transaction Record

Date Processed	Date of Transaction	Details	Amount As
11/08/06	10/08/06	AGL RETAIL ENERGY NORTH SYDNEY	\$137.85
12/08/06	11/08/06	OPTUS MOBILE AUTOPAY NORTH SYDNEY	\$22.00
12/08/06	11/08/06	OPTUS MOBILE AUTOPAY NORTH SYDNEY	\$55.00
16/08/06	15/08/06	WORLD VISION OF AUST BURWOOD EAST	\$39.00
17/08/06	17/08/06	CASH PAYMENT - THANK YOU MELBOURNE AU	\$160.00 CR
19/08/06	19/08/06	CHEQUE PAYMENT THANK YOU AU	\$127.35 CR
22/08/06	19/08/06	OFFICEWORKS 206 Brisbane	\$134.80
23/08/06	23/08/06	NRMA PBTM SYDNEY AU	\$25.09
23/08/06	22/08/06	OPTUS WIRELINE A/PAY NORTH SYDNEY	\$61.38
25/08/06	25/08/06	CASH PAYMENT - THANK YOU MELBOURNE AU	\$90.00 CR
26/08/06	23/08/06	WOOLWORTHS SYDNEY	\$20.00
29/08/06	28/08/06	NRMA SYDNEY	\$47.47
31/08/06	31/08/06	CASH PAYMENT THANK YOU MELBOURNE AU	\$175.00 CR
1/09/06	31/08/06	DAVID JONES SYDNEY	\$285.09
6/09/06	5/09/06	GIO/SML SYDNEY	\$49.34

Unauthorized or Unknown Transactions: To report an unauthorized or unknown transaction, please call 13 476. It is in your interest to report any unauthorized or unknown transactions immediately, as a delay in notification may limit the Aussie ability to investigate the transactions.

For your
record

Date Paid	Chq/Rec no.	Amount \$
-----------	-------------	-----------

bill 7: aussie visa – credit card monthly statement



Mr D Rose
41 Table Rd
Port Melden 2731

Statement for
Aussie Visa
ABN 473 685 291

Your Account Summary

Balance from Previous Statement	\$331.06 DR
Payments and other credits	\$341.00 CR
Purchases, cash advances	\$687.94 DR
Interest and other charge	\$0.00
Closing Balance	\$
Monthly Payment Due By 23 September 2005	\$
Minimum Payment	\$

Statement Period:	11 June 2006 to 9 July 2006
Account No:	6554 4122 9734
Credit Limit:	\$6,000
Available Credit:	
Page No:	1

Closing Balance Breakdown

Transaction Type	Annual % Rate	Daily % Rate	Balance
Purchase	16.650	0.04561	

Your Transaction Record

Date Processed	Date of Transaction	Details	Amount As
14/06/06	13/06/06	OPTUS AUTOPAY NORTH SYDNEY	\$9.00
14/06/06	13/06/06	OPTUS AUTOPAY NORTH SYDNEY	\$77.00
18/06/06	17/06/06	WOOLWORTHS SYDNEY	
21/06/06	20/06/06	OFFICEWORKS 206 BRISBANE	\$124.30
22/06/06	21/06/06	DAVID JONES SYDNEY	\$41.50
23/06/06	23/06/06	CASH PAYMENT THANK YOU AU	\$341.00 CR
29/06/06	28/06/06	COLES FYSHWICK	\$44.09
03/07/06	01/07/06	K MART MIRANDA	\$87.15
06/07/06	03/07/06	CAPT'N SNOOZE SYDNEY	\$143.50
06/07/06	04/07/06	COLES EXPRESS DUBBO	\$27.30
07/07/06	06/07/06	NRMA PBTM SYDNEY AU	\$31.40
09/07/06	08/07/06	PIZZA HUT DUBBO	\$25.60

Unauthorized or Unknown Transactions: To report an unauthorized or unknown transaction, please call 13 476. It is in your interest to report any unauthorized or unknown transactions immediately, as a delay in notification may limit the Aussie ability to investigate the transactions. Despatch here.

For your record

Date Paid	Chq/Rec no.	Amount \$
-----------	-------------	-----------

1-salaries

- (1) \$26 104 (2) \$39 312 (3) \$49 816 (4) \$23 296
 (5) \$25 948 (6) \$45 708 (7) \$25 740 (8) \$38 016
 (9) \$62 508 (10) \$1503.67 (11) \$3882.67
 (12) \$1950 (13) \$79 040 (14) \$3033.33 (15) \$4500
 (16) (i) \$1211 (ii) \$2422 (iii) \$5247.67 (17) \$4576
 (18) \$1971.67 (19) \$8991.67 (20) (ii) by \$1508
 (21) (i) by \$78 (22) (i) by \$2110 (23) (i) by \$11
 (24) (i) \$5.77 (25) (i) by \$125.50

2-wages

- (1) \$393.75 (2) \$347.80 (3) (i) \$336 (ii) \$459.20
 (iii) \$408.80 (4) (i) \$135.30 (ii) \$676.50 (iii) \$35 178
 (5) (i) \$178.20 (ii) \$891 (iii) \$46 332 (6) (i) \$58.50
 (ii) \$292.50 (iii) \$15 210 (7) Mon 8 hr, Tue 7 hr, Wed
 7 hr 15 min, Thu 9 hr, Fri 7½ hr, \$418.50 (8) \$11.35
 (9) \$8.70 (10) \$8.70 (11) \$27 378 (12) \$18 525
 (13) 36½ hr (14) \$9.60 (15) \$10.80

3-leave loading, overtime and special allowances

- (1) \$346.50 (2) \$510.30 (3) \$431.90 (4) \$366.10
 (5) \$478.80 (6) \$649.60 (7) \$3736.50 (8) \$5860.90
 (9) \$3412.20 (10) \$4112.50 (11) \$130.50 (12) \$110.25
 (13) \$141 (14) \$319.20 (15) \$215 (16) \$335.40
 (17) \$206.40 (18) \$395.60 (19) \$490.71 (20) \$785.50
 (21) \$677.72 (22) \$692.35 (23) \$717.68 (24) \$798.18
 (25) \$836.56 (26) \$884.12 (27) 7 hours

4-commission, royalties and piecework

- (1) \$918.75 (2) \$530 (3) \$293.75 (4) \$5760
 (5) \$69 500 (6) \$762.50 (7) \$43 000 (8) 5% (9) \$3620
 (10) \$4560 (11) \$630 (12) \$857.25 (13) \$173.88
 (14) \$10 131.75 (15) (i) \$2 003 300 (ii) \$1 048 950
 (iii) \$946 125 (iv) \$795 150 (v) \$1 597 842
 (16) \$630 (17) 9.5% (18) \$655.50, \$59.59 (19) \$163.20
 (20) \$370, \$9.74 (21) \$709.50 (22) (i) \$4514 (ii) \$8967
 (23) 30.5 m²

5-income from government allowances

- (1) (i) \$183.20 (ii) \$236 (iii) \$520.57 (2) (i) \$334.70
 (ii) \$62 (iii) \$563 (3) (i) \$438.50 (ii) \$236
 (iii) \$885.29 (4) (i) \$367.5 (ii) \$236 (iii) \$783.86
 (5) \$220.30 (6) \$220.30 (7) \$34 (8) \$0
 (9) \$188.30 (10) \$399.50 (11) \$233.10
 (12) \$499.70 (13) \$486.90 (14) (i) \$834.40
 (ii) \$820 (iii) \$551.60 (15) (i) \$293.80 (ii) \$2394.30

6-deductions from gross pay

- (1) \$544.72 (2) \$632.21 (3) \$995.11 (4) \$1040.95
 (5) \$607.42 (6) \$1062.85 (7) \$810.05 (8) \$1097.47
 (9) \$883.42 (10) \$30 750.40, \$2562.53 (11) \$927.57
 (12) \$408.62 (13) \$494.36 (14) \$705.34 (15) \$635.06
 (16) \$821.42 (17) \$874.32 (18) \$931.52 (19) \$1195.15

7-financial mathematics review 1

- (1) \$577.50 (2) \$543.75 (3) 39.5 hours (4) (i) \$151.90
 (ii) \$759.50 (iii) \$39 494 (5) \$1456.50 (6) (i) 8 (ii) 8
 (iii) 8¼ (iv) 8½ (v) 7½ (vi) \$31 101.98 (7) (i) \$25 246
 (ii) \$2103.83 (8) \$2028 (9) 4.5% (10) \$30 348
 (ii) by \$806.80 (11) \$631.92 (12) \$767.12

- (13) \$988.22 (14) \$33 550.92 (15) Sue by \$1952.2
 (16) \$3694.20 (17) \$500.25 (18) (i) \$577.20
 (ii) \$647.40 (iii) \$826.80 (iv) 45 hrs (19) \$652.80
 (20) \$879.37 (21) \$1911 (22) \$4512.50
 (23) \$16 390.50 (24) \$1897.56 (25) (i) \$183.20
 (ii) \$153.70 (26) (i) none (ii) \$36.10 (27) \$438.50
 (28) \$114.15 (29) \$491.30 (30) \$806.08

8-reading bills

- (1) Phone or Internet (2) 55.5% (3) 0.62 kL
 (4) 125 (5) 0.68 kL/day (6) 101c/kL (7) 42c/kL
 (8) 50% (9) w \$96.96 s \$20.16 (10) 11 250 L
 (11) 679 482 45 (12) 31/08/06 (13) \$285.09 (14) 4
 (15) No (16) 3 months (17) Residential Ad Valorem,
 Rivers Catchment Authority, Environmental Levy
 (18) \$210 000 (19) 9% pa, daily (20) \$531.30, \$15
 (21) \$191.27 (22) 46 38 74 49 (23) yes (24) 15/6/2006
 (25) 17 kwh (26) 1368.9 kg (27) \$197.00 (28) 5.6%

9-budgets

- (1) (i) \$35 880 (ii) \$550 (iii) \$7800 (iv) \$1120
 (v) \$660 (vi) \$582.40 (vii) \$520 (viii) \$1560
 (ix) \$1040, 3 years (2) \$2534 (3) \$1343 (4) yes, \$63/
 week, \$782 left (5) No, expenditure \$174 more than
 income (6) (i) \$83.44 (ii) \$362

10-simple interest

- (1) \$877.50 (2) \$2850 (3) \$287.85 (4) \$1230.88
 (5) \$5613.30 (6) \$1191.45 (7) \$326.25 (8) 13 446.40
 (9) \$4780 (10) 9.5% pa (11) 9 yrs (12) 0.9%, 5.4%,
 2.7%, 14.4%, 7.2%, 3.6%, 9%, 0.75%, 2.25%, 16.14%,
 1.345%, 8.07% (13) \$131.25 (14) (i) \$658.35
 (ii) \$804.65 (15) (i) \$73.08, (ii) \$156.60
 (iii) \$217.44 (16) 5 mths (17) 9% (18) 11 mths

11-compound interest

- (1) \$1400, \$140, \$1540, \$154, \$1694, Int. \$294 (2) \$2500,
 \$225, \$2725, \$245.25, \$2970.25, \$267.32, \$3237.57 Int.
 \$737.57 (3) \$4233.58 (4) \$3050.66 (5) \$10 232.55
 (6) (i) \$9289.92 (ii) \$13 968.61 (iii) \$59 951.48
 (7) (i) \$10 098 (ii) \$16 948.42, \$6850.42 (8) \$3732.08
 (9) \$6520.41 (10) \$8305.88 (11) \$10 364.98
 (12) \$12 025.42 (13) \$8196.68 (14) Pam by \$382.59

12-income tax and Medicare levy

- (1) Nil (2) \$1177.50 (3) \$14 262 (4) \$26 018
 (5) \$52 080 (6) (i) \$33 800 (ii) \$5490 (iii) \$544.42
 (7) \$46.98 (8) \$227.38 (9) Nil (10) Nil (11) \$264
 (12) \$268.68 (13) \$325.50 (14) \$1098.75
 (15) \$18 894.25 (16) \$29 686.75 (17) \$56 405.25
 (18) \$43 048.36 (19) \$2037.94 (20) (i) \$38 945
 (ii) \$7033.50 (iii) \$584.18 (iv) \$31 327.32
 (21) \$31 649.96, \$608.65 (22) \$2154.39

13-income tax and deductions

- (1) (i) \$14 380 (ii) \$1257 (iii) Nil (iv) Refund \$33
 (2) (i) \$16 322 (ii) \$1548.30 (iii) \$7.60 (iv) Refund
 \$64.10 (3) (i) \$26 354 (ii) \$3256.20 (iii) \$395.31
 (iv) Owe \$351.51 (4) (i) \$55 930 (ii) \$12 129
 (iii) \$838.95 (iv) Refund \$332.05 (5) (i) \$75 610
 (ii) \$18 094 (iii) \$1134.15 (iv) Owe \$128.15 (6) Refund
 \$1113.21 (7) Owe \$5243.64 (8) Refund \$24.25

14-financial mathematics review 2

- (1) \$11 per week (2) (i) yes (ii) \$16.34 (iii) \$30.32
 (iv) \$24.51 (v) approx \$50 (vi) 0294731841 (vii) \$1.55
 (3) \$5400 (4) \$967.50 (5) \$2809.33 (6) \$16 625
 (7) \$11 683.02 (8) \$11 974.14 (9) \$9604.89
 (10) \$7626.22 (11) \$80 496 (12) nil (13) \$1722
 (14) \$7035 (15) nil (16) \$99.20 (17) \$712.50
 (18) (i) 840.18 (ii) \$1178.43 (iii) \$1077.40
 (iv) \$284.04 (19) (i) \$6375 (ii) \$551.25
 (20) (i) \$47 303 (ii) \$9540.90 (iii) \$37 052.55
 (21) To pay \$383.34

15-GST and VAT

- (1) \$89 (2) 15c (3) \$2585 (4) 10c (5) \$5.60 (6) 45c
 (7) (i) \$84 (ii) \$100.80 (iii) \$75.60 (iv) 73.50
 (v) \$82.32 (vi) \$67.20 (8) 20c (9) \$3586 (10) \$67.50
 (11) \$175.60 (12) 75c (13) \$58 (14) (i) \$323.69
 (ii) \$359 (iii) \$294.16 (iv) \$247.59 (15) \$123.36
 (16) \$89.65 (17) \$273.45 (18) \$224.16 (19) \$117.54

16-flat rate loans

- (1) (i) \$4687.50 (ii) \$17 187.50 (iii) \$286.46
 (2) (i) \$1260 (ii) \$5460 (iii) \$151.67 (3) (i) \$3744
 (ii) \$8944 (iii) \$124.22 (4) (i) \$9450 (ii) \$25 200
 (iii) \$350 (5) (i) \$1974 (ii) \$6674 (iii) \$139.04
 (6) (i) \$7234.50 (ii) \$14 654.50 (iii) \$122.12
 (7) (i) \$11 897.60 (ii) \$22 297.60 (iii) \$232.27
 (8) (i) \$2223.46 (ii) \$7099.46 (iii) \$147.91
 (9) (i) \$11 609.57 (ii) \$24 079.57 (iii) \$286.66
 (10) (i) \$3860.52 (ii) \$9930.52 (iii) \$137.92
 (11) \$255.31 (12) \$197.56 (13) Option iv, \$335.22
 (14) 7.5% p.a. (15) 15% p.a. (16) 5 yrs (17) 7% p.a.
 (18) 10% p.a. (19) 3½ yrs (20) \$309.38
 (21) \$1096.88 (22) \$5699.71 (23) 6.5% p.a.
 (24) \$73.17 (25) (i) 8 years, 4 months
 (ii) 7½ years (26) (i) \$544.43 (ii) 89.26

17-future value formula

- (1) (i) \$4234.74 (ii) \$3885.09 (iii) 3000 (1.09)²
 (iv) \$3564.30 (v) 3000 (1.09)¹ (vi) \$3270
 (vii) 3000 (1.09)⁰ (viii) \$3000 (ix) \$17 954.13
 (2) \$30 622.31 (3) \$11 274.19 (4) \$27 527.28
 (5) \$51 741.52 (6) \$6811.88 (7) \$129 888.80
 (8) \$29 425.81 (9) \$28 502.11 (10) \$55 974.55
 (11) \$11 828.75 (12) \$8259.37 (13) No, \$3291.49
 (14) \$49.74 (15) Bobby by \$270.51
 (16) (ii) by \$5892.79 (17) \$32 698.81

18-present value of an annuity

- (1) \$14 299.62 (2) \$26 840.33 (3) \$24 556.45
 (4) \$33 943.56 (5) \$27 880.21 (6) \$46 562.06
 (7) \$34 760.89 (8) \$30 320.05 (9) \$6144.57
 (10) \$15 980.51 (11) \$5587.07

19-problems involving present value of an annuity

- (1) (i) 300 000 (ii) 60 (iii) 10.8, 0.009 (iv) 300 000
 (v) 60 (vi) 0.009 (vii) 1.009 (viii) 300 000
 (ix) 46.204 71 (x) 6492.84 (xi) \$389 570.40,
 \$89 570.40 (2) (i) 48 (ii) 0.01 (iii) 1000 (iv) 1000
 (v) 1.01 (vi) 0.01 (vii) 1.01 (viii) \$37 974

- (3) \$1263.87 (4) \$2201.65 (5) \$155.07 (6) \$1187.88
 (7) \$1672.82 (8) (i) \$1799.45, \$431 868.46, \$231 868
 (ii) \$216.76, \$39 016.80, \$18016.80 (9) (i) \$1366.15
 (ii) \$1516.32 (iii) \$33 138.48 (10) \$16 826.88
 (11) \$68 390.30 (12) \$40 648.56 (13) \$31 035.34
 (14) \$26 063.27 (15) (i) \$311 205.87 (ii) \$22 228.99
 (16) \$191 358.35 (17) \$250 469.83

20-financial mathematics review 3

- (1) (i) \$348.50 (ii) \$1535 (iii) \$31 (iv) \$9.50
 (2) (i) \$435.63 (ii) \$1918.75 (iii) \$38.75 (iv) \$11.88
 (3) \$49 918 (4) (i) \$185.80 (ii) \$18.33 (5) (i) \$433.13
 (ii) \$206.25 (6) \$4284 (7) \$414.75 (8) 8% p.a.
 (9) \$37 693.53 (10) \$724 904.44 (11) \$102 856.74
 (12) \$1496.43 (13) \$13 800.65 (14) yes, \$97 500 over

21-shares buying and selling

- (1) 2.88% (2) 2.20% (3) 5.36% (4) 2.10% (5) 3.13%
 (6) 6.48% (7) 3.97% (8) 3.44% (9) 92.97c
 (10) 51.00c (11) 34.00c (12) 18.48c (13) 21.06c
 (14) 41.23c (15) 114.26c (16) (i) \$19 224 (ii) \$161.10
 (iii) \$24 596.10 (17) (i) \$6735 (ii) \$360c (iii) \$5910
 (18) Katherine by \$332.88 (19) (i) \$75 232.43
 (ii) \$85 352.77 (iii) Profit \$10 120.34 (20) (i) \$96 693.48
 (ii) \$89 267.47 (iii) Loss \$7426.01 (21) (i) \$18 842.82
 (ii) \$13 668.32 (iii) Loss \$5174.50 (22) (i) 1191
 (ii) \$9239.38

22-guess and check to determine the number of periods

- (1) 240 (2) 300 (3) 180 (4) 120 (5) 195 (6) 294
 (7) 122 (8) 144 (9) 126 (10) 140 (11) 240 (12)
 300 (13) 360 (14) 217

23-depreciation

- (1) \$1500 (2) \$4790 (3) \$310 (4) \$1550 (5) \$8100
 (6) \$17 527.54 (7) \$4590.43 (8) \$3986.72 (9) \$864.00
 (10) \$15 045.92 (11) Fred \$34 300, \$11 765, \$5765. Dave
 \$50 000, \$20 000, \$0. Approx 6 yr 3 mth. (12) 30%,
 \$11 662 (13) \$3000 (14) \$5832 (15) \$62 500

24-credit cards

- (1) \$653.33 (2) yes (3) Australian dollars, credit
 (4) \$644.83 (5) 23/12/06 (6) \$16.12 (8) \$35.87
 (9) \$6000 (11) \$277.34 (12) \$877.02 (13) \$612.01
 (14) Credit Limit minus closing balance (cents taken up
 to next dollar) (15) 134 766 (16) 0.04561% (17) \$15.81
 (18) Debit (19) \$678.00 (20) \$16.95
 (21) \$5322 (22) \$77.10 (23) Yes (24) 1
 (25) 29 (26) \$7.99 (27) 0.0466 (28) \$15.57
 (29) \$61.15 (30) \$34.20 (31) \$34.67

25-general review

- (1) \$72 (2) \$10.92 (3) 8750 (4)(i) \$2625
 (ii) \$1338.80 (iii) 7th (5) 9.42c (6) \$32.21
 (7) (i) \$12 566.45 (ii) \$3446.30 (8) \$15.30
 (9) \$702.50 (10) \$197 208.15 (11) \$2049.31, \$245 916.61,
 \$95 916.61 (12) \$48 221.74

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